

Survival Regression Analysis

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Learning Objectives

- Visualize cumulative incidence of cancer deaths and other deaths in cancer patients.
- Use Cox proportional hazards models to compare mortality rates among different demographic groups.
- Calculate, visualize and interpret time-varying hazard ratios.

1. Import and manage data

```
data <- read.csv("melanoma_survival.csv", header=TRUE)
```

```
# Age
```

```
table(data$Age.recode.with..1.year olds.and.90.)
```

15-19 years	20-24 years	25-29 years	30-34 years	35-39 years	40-44 years
1412	3908	7302	11183	14862	19325
45-49 years	50-54 years	55-59 years	60-64 years	65-69 years	70-74 years
26097	33805	41309	47299	50594	47770
75-79 years	80-84 years	85-89 years	90+ years		
42301	34539	22827	12385		

```
data$age <- dplyr::case_when(
  data[[1]] %in% c("15-19 years", "20-24 years", "25-29 years", "30-34 years",
    "35-39 years", "40-44 years", "45-49 years") ~ "<50",
  data[[1]] %in% c("50-54 years", "55-59 years") ~ "50-59",
  data[[1]] %in% c("60-64 years", "65-69 years") ~ "60-69",
  data[[1]] %in% c("70-74 years", "75-79 years", "80-84 years", "85-89 years", "90+ years") ~ "70+",
  TRUE ~ NA_character_
)
data$age <- factor(data$age, levels = c("70+", "<50", "50-59", "60-69"))
```

```
# Race
```

```
colnames(data)[2] <- "race"
```

```
data$race <- factor(data$race)
```

```
data$race <- relevel(data$race, ref = "Non-Hispanic White")
```

```
# Sex
```

```
colnames(data)[3] <- "sex"
```

```
data$sex <- factor(data$sex, levels = c("Female", "Male"))
```

```

# Create Head_and_Neck classification and update subsite factor levels
# Rename column 5 to "subsite"
colnames(data)[5] = "subsite"

# Recode subsite into broader categories
data <- data %>%
  mutate(
    subsite = case_when(
      subsite %in% c("C44.1-Eyelid", "C44.2-External ear",
                    "C44.3-Skin other/unspec parts of face",
                    "C44.4-Skin of scalp and neck") ~ "Head and Neck",
      subsite %in% c("C44.5-Trunk") ~ "Trunk",
      subsite %in% c("C44.6-Upper limb") ~ "Arm",
      subsite %in% c("C44.7-Lower limb") ~ "Leg",
      TRUE ~ subsite
    ),
    subsite = factor(subsite, levels = c("Head and Neck", "Trunk", "Arm", "Leg"))
  )

# Year of diagnosis
data$yod <- ifelse(data[[4]] < 2010, "2004-2009", "2010-2022")
data$yod <- factor(data$yod, levels = c("2004-2009", "2010-2022"))

# Stage
colnames(data)[6] = "stage"
table(data$stage)

```

```

Distant site(s)/node(s) involved
8504
In situ
47
Localized only
345988
Regional by both direct extension and lymph node involvement
4340
Regional by direct extension only
11179
Regional lymph nodes involved only
19852
Unknown/unstaged/unspecified/DCO
27008

```

```

data$stage <- dplyr::case_when(
  data$stage %in% c("Blank(s)", "In situ", "Unknown/unstaged/unspecified/DCO") ~ "Unstaged",
  data$stage %in% c("Regional by both direct extension and lymph node involvement",
                    "Regional by direct extension only",
                    "Regional lymph nodes involved only") ~ "Regional",
  data$stage == "Distant site(s)/node(s) involved" ~ "Distant",
  data$stage == "Localized only" ~ "Localized",

```

```

TRUE ~ data$stage
)
data$stage <- factor(data$stage, levels = c("Localized","Regional","Distant","Unstaged"))

# Survival time
colnames(data)[9] <- "time"
data$time <- as.numeric(data$time)
data$time <- data$time / 12 # convert months to years

# Survival event
data$event <- dplyr::case_when(
  data[[7]] == "Dead (attributable to this cancer dx)" ~ "Cancer death",
  data[[8]] == "Dead (attributable to causes other than this cancer dx)" ~ "Other death",
  data[[7]] == "Alive or dead of other cause" & data[[8]] == "Alive or dead due to cancer" ~ "Alive",
  TRUE ~ NA_character_
)

```

2. Plot cumulative incidence using the survival package

```

# Create a multistate (competing risks) survival object with no covariates (~1)
sf_HN_MS <- survfit(Surv(time=data$time,event=data$event,type="mstate")~1)
DATA <- matrix(NA,nrow=3*length(sf_HN_MS$time),ncol=3)

# Time for each of "Cancer death", "other death", and "alive"
DATA[,1] <- rep(sf_HN_MS$time,3)

# Probabilities are cumulative. These are where "Cancer death", "other death", and "alive" each start
DATA[,2] <- c(rep(0,length(sf_HN_MS$time)),sf_HN_MS$pstate[,2],sf_HN_MS$pstate[,2]+sf_HN_MS$pstate[,3])

# Probabilities are cumulative. These are where "Cancer death", "other death", and "alive" each end
DATA[,3] <- c(sf_HN_MS$pstate[,2],sf_HN_MS$pstate[,2]+sf_HN_MS$pstate[,3],rep(1,length(sf_HN_MS$time)))
DATA <- as.data.frame(DATA)

colnames(DATA) <- c("x","ymin","ymax")
DATA$state <- c(rep("Cancer Death",length(sf_HN_MS$time)),rep("Other death",length(sf_HN_MS$time)),
  rep("Alive",length(sf_HN_MS$time)))
DATA$state <- factor(DATA$state,levels=c("Cancer Death","Other death","Alive"))

p0 <- ggplot(data = DATA, aes(x = x, y = ymin, ymin = ymin, ymax = ymax, fill = state)) +
  scale_fill_manual(values=c("gray40", "gray2", "grey60"), guide_legend(title = "Age at diagnosis")) +
  geom_ribbon(aes(fill = state,linetype = NA), alpha = 1, show.legend = FALSE) +
  theme_bw(base_size = 8) + labs(x = 'Years since diagnosis', y = 'Fraction') +
  theme(axis.title.y = element_text(vjust=1.5))+
  theme(legend.position.inside = c(0.3,0.8),legend.key = element_blank(),
    legend.background = element_blank(), legend.text = element_text(size=7),
    legend.key.height = unit(0.05,"inch")) +
  theme(panel.border = element_blank(),
    panel.background = element_blank(),

```

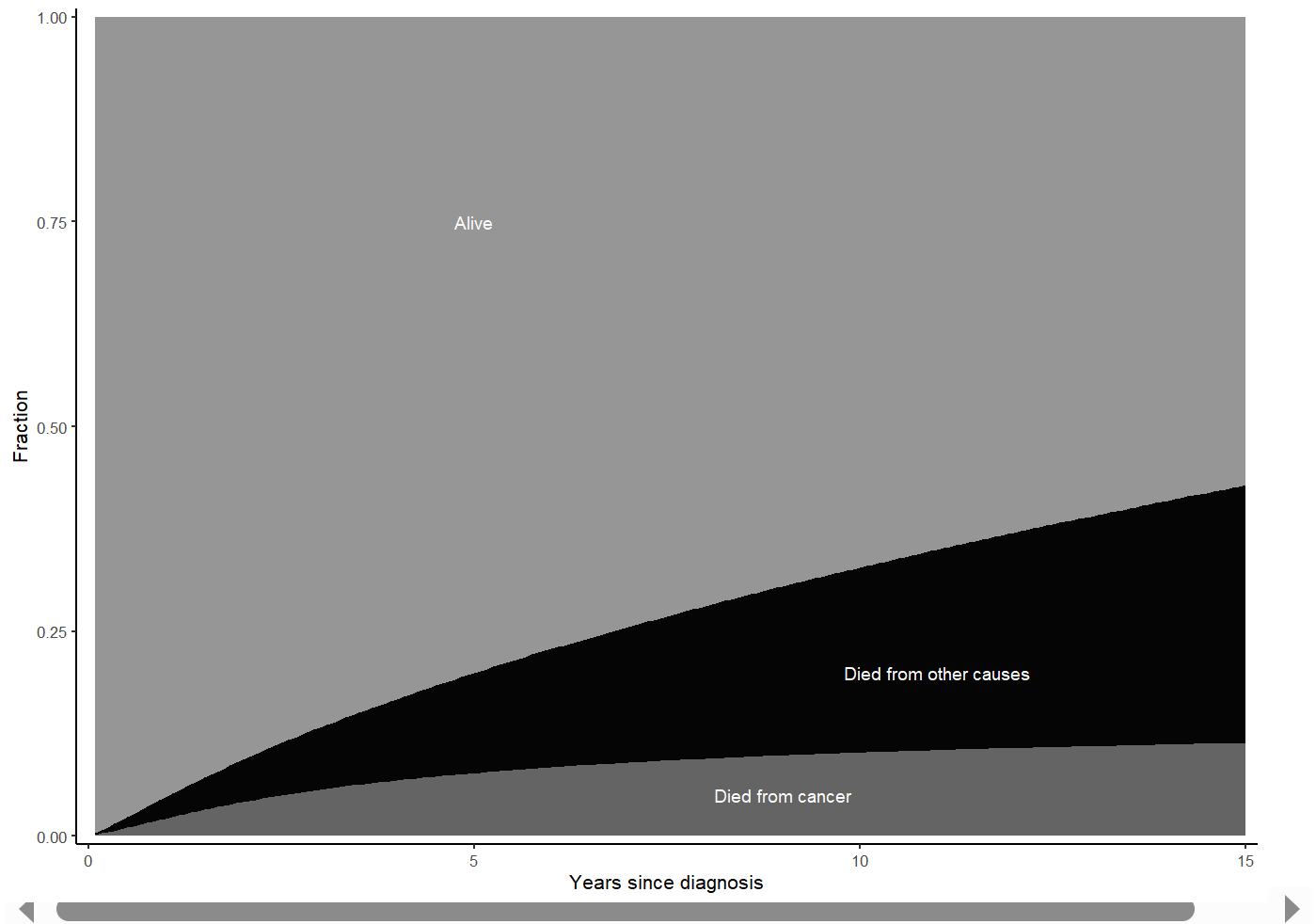
```

panel.grid.major = element_blank(),
panel.grid.minor = element_blank(),
axis.line.x = element_line(color = "black", linewidth = 0.4),
axis.line.y = element_line(color = "black", linewidth = 0.4))+
annotate("text", x = c(5, 9, 11), y = c(0.75, 0.05, 0.2), label = c("Alive", "Died from cancer",
                                                                    "Died from other causes"),
          color = "White", size = 2.5) +
scale_x_continuous(limits = c(0, 15), expand = (c(0.01, 0.01)), breaks = c(0, 5, 10, 15, 20)) +
scale_y_continuous(limits = c(0, 1), expand = c(0.005, 0.005))

```

p0

Warning: Removed 141 rows containing missing values or values outside the scale range (``geom_ribbon()``).



3. Plotting cumulative incidence by covariate

Here, we create a `survfit` object with cumulative incidence stratified by the levels of our variables of interest. In order to extract information for plotting from the `survfit` objects, we need to know the order of the levels and how many rows are associated with each level.

Let's start by looking at sex and race & ethnicity.

```
data$time <- as.numeric(as.character(data$time))

# Cumulative incidence of cancer death by sex
sf_sex <- survfit(Surv(time=data$time,event=data$event,type="mstate")~data$sex)

# Cumulative incidence of cancer death by race and ethnicity
sf_race_eth <- survfit(Surv(time=data$time,event=data$event,type="mstate")~data$race)

# Cumulative incidence of cancer death by subsite
sf_subsite <- survfit(Surv(time=data$time,event=data$event,type="mstate")~data$subsite)

# Cumulative incidence of cancer death by age
sf_age <- survfit(Surv(time=data$time,event=data$event,type="mstate")~data$age)

# Check the strata of each model
sf_sex$strata
```

```
data$sex=Female    data$sex=Male
          227          227
```

```
sf_race_eth$strata
```

```
          data$race=Non-Hispanic White
                      227
          data$race=Hispanic (All Races)
                      227
data$race=Non-Hispanic American Indian/Alaska Native
                      213
          data$race=Non-Hispanic Asian or Pacific Islander
                      227
          data$race=Non-Hispanic Black
                      227
          data$race=Non-Hispanic Unknown Race
                      227
```

```
sf_subsite$strata
```

```
data$subsite=Head and Neck    data$subsite=Trunk
                      227          227
          data$subsite=Arm    data$subsite=Leg
                      227          227
```

```
sf_age$strata
```

```
data$age=70+    data$age=<50    data$age=50-59    data$age=60-69
          227          227          227          227
```

In the example data, most groups have information across all possible survival months, but that isn't true for small subgroups (in the example data, Non-Hispanic American Indian/Alaskan Native).

```
# Plot cumulative incidence of cancer death by sex
# For ease of plotting for other covariates, use generic plot code after setting
df_sex <- sf_sex
mycolors <- magma

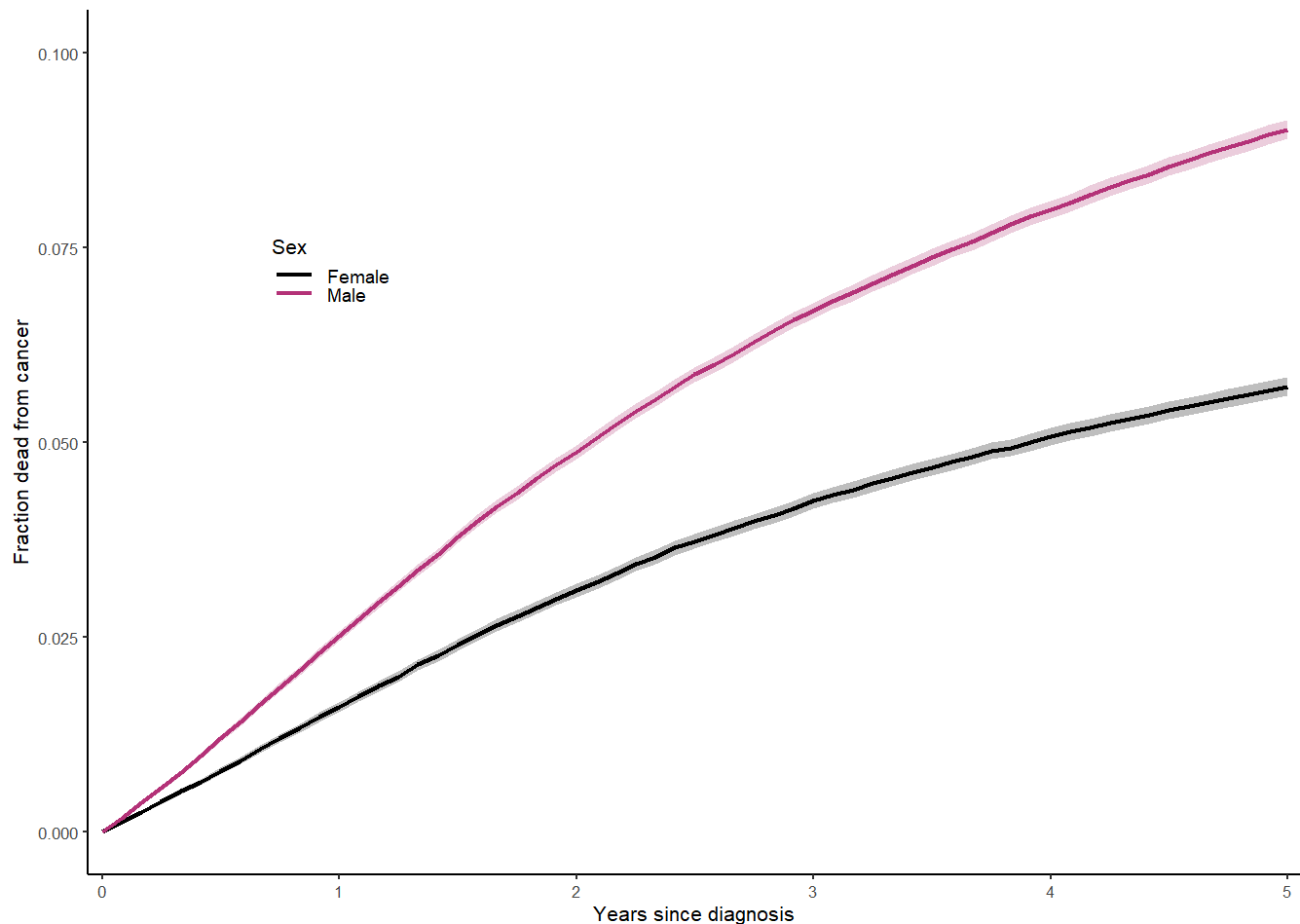
# Create a data matrix with the correct number of rows:
# A 0 for each group plus the sum of the number of rows for each subgroup
DATA_ci_sex <- matrix(NA,nrow=(length(df_sex$strata)+sum(df_sex$strata)),ncol=4)
# x
DATA_ci_sex[,1] <- c(0,df_sex$time[1:df_sex$strata[1]],
                    0, df_sex$time[(1 + df_sex$strata[1]):(sum(df_sex$strata[1:2]))])
# y
DATA_ci_sex[,2] <- c(0,df_sex$pstate[1:df_sex$strata[1],2],
                    0, df_sex$pstate[(1 + df_sex$strata[1]):(sum(df_sex$strata[1:2])),2])
# 95% CI (lower bound)
DATA_ci_sex[,3] <- c(0,df_sex$lower[1:df_sex$strata[1],2],
                    0, df_sex$lower[(1 + df_sex$strata[1]):(sum(df_sex$strata[1:2])),2])
# 95% CI (upper bound)
DATA_ci_sex[,4] <- c(0,df_sex$upper[1:df_sex$strata[1],2],
                    0, df_sex$upper[(1 + df_sex$strata[1]):(sum(df_sex$strata[1:2])),2])
DATA_ci_sex <- as.data.frame(DATA_ci_sex)
colnames(DATA_ci_sex) <- c("x","y","ymin","ymax")
breaks <- which(c(DATA_ci_sex$x,0)==0)[-1]
DATA_ci_sex$sex <- c(rep("Female",df_sex$strata[1] + 1),rep("Male",df_sex$strata[2] + 1))
DATA_ci_sex$sex <- factor(DATA_ci_sex$sex,levels=levels(data$sex))

# In this plot, we have two groups, so it gets busy to also show other
# cause deaths. We drop them from this plot.
plot_sex <- ggplot(data = DATA_ci_sex,aes(x = x, y = y, ymin = ymin, ymax = ymax, color = sex)) +
  scale_color_manual(values = mycolors(3)[-3],name = "Sex") +
  geom_ribbon(aes(fill = sex, linetype = NA), alpha = 0.25, show.legend=FALSE) +
  geom_line(linewidth = 0.75) +
  scale_fill_manual(guide = "none", values = mycolors(3)[-3]) +
  theme_bw(base_size = 8) + labs(x = 'Years since diagnosis', y = 'Fraction dead from cancer') +
  theme(axis.title.y = element_text(vjust=1.5)) +
  theme(legend.position = "inside",
        legend.position.inside = c(0.2,0.7), legend.key = element_blank(),
        legend.background = element_blank(), legend.text = element_text(size = 7),
        legend.key.height = unit(0.05,"inch")) +
  theme(panel.border = element_blank(),
        panel.background = element_blank(),
        panel.grid.major = element_blank(),
        panel.grid.minor = element_blank(),
        axis.line.x = element_line(color = "black", linewidth = 0.4),
        axis.line.y = element_line(color = "black", linewidth = 0.4)) +
  scale_x_continuous(limits = c(0, 5), expand = (c(0.01, 0.01))) +
  scale_y_continuous(limits=c(0, 0.1), expand =c(0.005, 0.005))
```

```
plot_sex
```

Warning: Removed 334 rows containing missing values or values outside the scale range (`geom\_ribbon()`).

Warning: Removed 334 rows containing missing values or values outside the scale range (`geom\_line()`).



```
# Plot cumulative incidence of cancer death by race and ethnicity
# For ease of plotting for other covariates, use generic plot code after setting
df_re <- sf_race_eth
# Create a data matrix with the correct number of rows:
# A 0 for each group plus the sum of the number of rows for each subgroup.
# Drop "unknown" race.
DATA_ci_re=matrix(NA, nrow=(5 + sum(df_re$strata[1:5])), ncol=4)
# x
DATA_ci_re[,1]=c(0, df_re$time[1:df_re$strata[1]],
  0, df_re$time[(1 + df_re$strata[1]):(sum(df_re$strata[1:2]))],
  0, df_re$time[(1 + sum(df_re$strata[1:2])):(sum(df_re$strata[1:3]))],
  0, df_re$time[(1 + sum(df_re$strata[1:3])):(sum(df_re$strata[1:4]))],
  0, df_re$time[(1 + sum(df_re$strata[1:4])):(sum(df_re$strata[1:5]))])
# y
```

```

DATA_ci_re[,2]=c(0, df_re$pstate[1:df_re$strata[1],2],
                0, df_re$pstate[(1 + df_re$strata[1]):(sum(df_re$strata[1:2])),2],
                0, df_re$pstate[(1 + sum(df_re$strata[1:2])):(sum(df_re$strata[1:3])),2],
                0, df_re$pstate[(1 + sum(df_re$strata[1:3])):(sum(df_re$strata[1:4])),2],
                0, df_re$pstate[(1 + sum(df_re$strata[1:4])):(sum(df_re$strata[1:5])),2])
# 95% CI (lower bound)
DATA_ci_re[,3]=c(0,df_re$lower[1:df_re$strata[1],2],
                0, df_re$lower[(1 + df_re$strata[1]):(sum(df_re$strata[1:2])),2],
                0, df_re$lower[(1 + sum(df_re$strata[1:2])):(sum(df_re$strata[1:3])),2],
                0, df_re$lower[(1 + sum(df_re$strata[1:3])):(sum(df_re$strata[1:4])),2],
                0, df_re$lower[(1 + sum(df_re$strata[1:4])):(sum(df_re$strata[1:5])),2])
# 95% CI (upper bound)
DATA_ci_re[,4]=c(0,df_re$upper[1:df_re$strata[1],2],
                0, df_re$upper[(1 + df_re$strata[1]):(sum(df_re$strata[1:2])),2],
                0, df_re$upper[(1 + sum(df_re$strata[1:2])):(sum(df_re$strata[1:3])),2],
                0, df_re$upper[(1 + sum(df_re$strata[1:3])):(sum(df_re$strata[1:4])),2],
                0, df_re$upper[(1 + sum(df_re$strata[1:4])):(sum(df_re$strata[1:5])),2])

DATA_ci_re <- as.data.frame(DATA_ci_re)
colnames(DATA_ci_re) <- c("x","y","ymin","ymax")
breaks <- which(c(DATA_ci_re$x,0) == 0)[-1]
DATA_ci_re$race_eth <- c(rep("Non-Hispanic White", 1 + df_re$strata[1]),
                        rep("Hispanic (All Races)", 1 + df_re$strata[2]),
                        rep("Non-Hispanic American Indian/Alaska Native", 1 + df_re$strata[3]),
                        rep("Non-Hispanic Asian or Pacific Islander", 1 + df_re$strata[4]),
                        rep("Non-Hispanic Black", 1 + df_re$strata[5]))
DATA_ci_re$race_eth=factor(DATA_ci_re$race_eth)

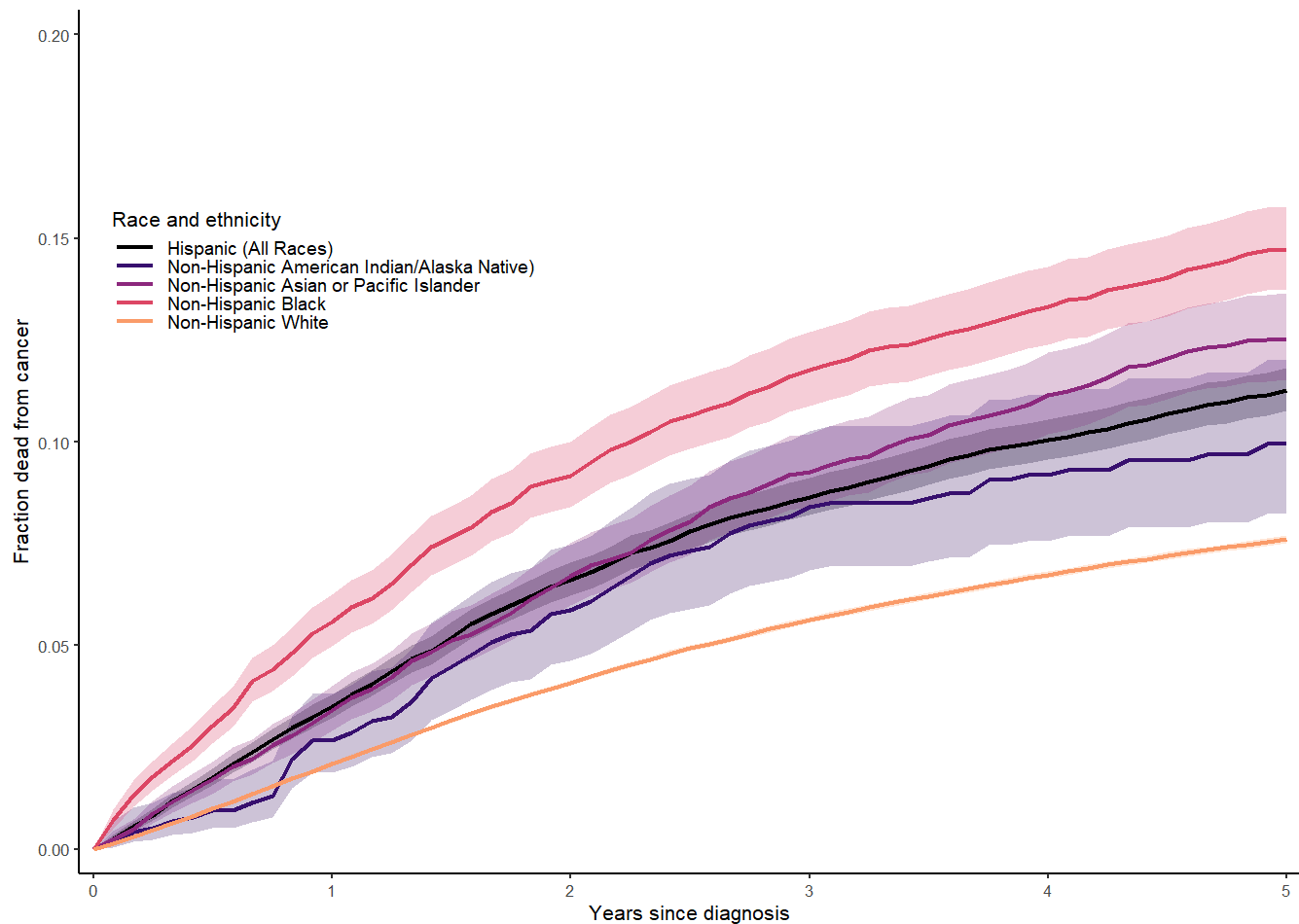
# In this plot, we have two groups, so it gets busy to also show other
# cause deaths. We drop them from this plot.
plot_race_eth <- ggplot(data=DATA_ci_re,aes(x = x, y = y, ymin = ymin, ymax = ymax, color = race_eth)) +
  scale_color_manual(values = mycolors(6)[-6], name = "Race and ethnicity") +
  geom_ribbon(aes(fill = race_eth,linetype = NA), alpha = 0.25, show.legend = FALSE) +
  geom_line(linewidth = 0.75) +
  scale_fill_manual(guide="none",values=mycolors(6)[-6])+
  theme_bw(base_size = 8) + labs(x = "Years since diagnosis", y = "Fraction dead from cancer") +
  theme(axis.title.y = element_text(vjust = 1.5)) +
  theme(legend.position = "inside",
        legend.position.inside = c(0.2,0.7),legend.key = element_blank(),
        legend.background = element_blank(),legend.text = element_text(size=7),
        legend.key.height = unit(0.05,"inch")) +
  theme(panel.border = element_blank(),
        panel.background = element_blank(),
        panel.grid.major = element_blank(),
        panel.grid.minor = element_blank(),
        axis.line.x = element_line(color = "black",linewidth=0.4),
        axis.line.y = element_line(color = "black",linewidth=0.4)) +
  scale_x_continuous(limits = c(0,5), expand = (c(0.01,0.01))) +
  scale_y_continuous(limits = c(0,.2), expand = c(0.005,0.005))

```

```
plot_race_eth
```

Warning: Removed 821 rows containing missing values or values outside the scale range (``geom_ribbon()``).

Warning: Removed 821 rows containing missing values or values outside the scale range (``geom_line()``).



```
# Plot cumulative incidence of cancer death by subsite
# For ease of plotting for other covariates, use generic plot code after setting
df_subsite <- sf_subsite
# Create a data matrix with the correct number of rows:
# A 0 for each group plus the sum of the number of rows for each subgroup.
# Drop "unknown" race.
DATA_ci_subsite=matrix(NA, nrow=(4 + sum(df_subsite$strata[1:4])), ncol = 4)
# x
DATA_ci_subsite[,1] <- c(0, df_subsite$time[1:df_subsite$strata[1]],
  0, df_subsite$time[(1 + df_subsite$strata[1]):(sum(df_subsite$strata[1:2]))],
  0, df_subsite$time[(1 + sum(df_subsite$strata[1:2])):(sum(df_subsite$strata[1:3]))],
  0, df_subsite$time[(1 + sum(df_subsite$strata[1:3])):(sum(df_subsite$strata[1:4]))])
# y
```

```

DATA_ci_subsite[,2] <- c(0,df_subsite$pstate[1:df_subsite$strata[1],2],
  0, df_subsite$pstate[(1 + df_subsite$strata[1]):(sum(df_subsite$strata[1:2])),2],
  0, df_subsite$pstate[(1 + sum(df_subsite$strata[1:2])):(sum(df_subsite$strata[1:3])),2],
  0, df_subsite$pstate[(1 + sum(df_subsite$strata[1:3])):(sum(df_subsite$strata[1:4])),2]
# 95% CI (lower bound)
DATA_ci_subsite[,3] <- c(0,df_subsite$lower[1:df_subsite$strata[1],2],
  0, df_subsite$lower[(1 + df_subsite$strata[1]):(sum(df_subsite$strata[1:2])),2],
  0, df_subsite$lower[(1 + sum(df_subsite$strata[1:2])):(sum(df_subsite$strata[1:3])),2],
  0, df_subsite$lower[(1 + sum(df_subsite$strata[1:3])):(sum(df_subsite$strata[1:4])),2]
# 95% CI (upper bound)
DATA_ci_subsite[,4] <- c(0,df_subsite$upper[1:df_subsite$strata[1],2],
  0, df_subsite$upper[(1 + df_subsite$strata[1]):(sum(df_subsite$strata[1:2])),2],
  0, df_subsite$upper[(1 + sum(df_subsite$strata[1:2])):(sum(df_subsite$strata[1:3])),2],
  0, df_subsite$upper[(1 + sum(df_subsite$strata[1:3])):(sum(df_subsite$strata[1:4])),2]
DATA_ci_subsite <- as.data.frame(DATA_ci_subsite)
colnames(DATA_ci_subsite) <- c("x","y","ymin","ymax")
breaks <- which(c(DATA_ci_subsite$x,0) == 0)[-1]
DATA_ci_subsite$subsite <- c(rep("Head and Neck", 1 + df_subsite$strata[1]),
  rep("Trunk", 1 + df_subsite$strata[2]),
  rep("Arm", 1 + df_subsite$strata[3]),
  rep("Leg", 1 + df_subsite$strata[4]))
DATA_ci_subsite$subsite = factor(DATA_ci_subsite$subsite)

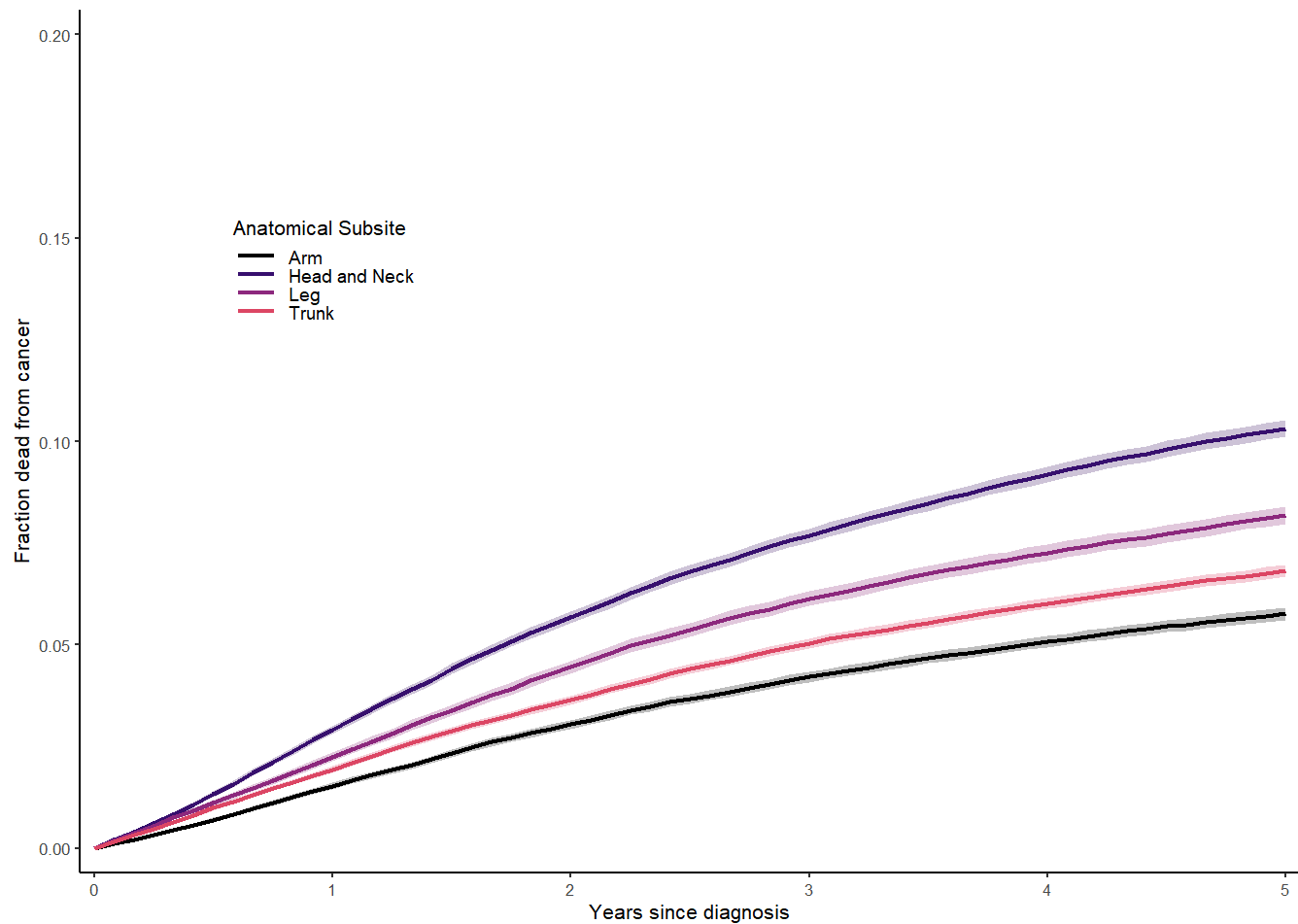
# In this plot, we have two groups, so it gets busy to also show other
# cause deaths. We drop them from this plot.
plot_subsite <- ggplot(data = DATA_ci_subsite, aes(x = x, y = y, ymin = ymin, ymax = ymax, color =
  scale_color_manual(values = mycolors(6)[-6],name = "Anatomical Subsite") +
  geom_ribbon(aes(fill = subsite, linetype = NA), alpha = 0.25, show.legend = FALSE) +
  geom_line(linewidth = 0.75) +
  scale_fill_manual(guide="none",values=mycolors(6)[-6]) +
  theme_bw(base_size = 8) + labs(x='Years since diagnosis', y='Fraction dead from cancer') +
  theme(axis.title.y=element_text(vjust=1.5)) +
  theme(legend.position = "inside",
    legend.position.inside = c(0.2,0.7), legend.key = element_blank(),
    legend.background = element_blank(), legend.text = element_text(size = 7),
    legend.key.height = unit(0.05,"inch")) +
  theme(panel.border = element_blank(),
    panel.background = element_blank(),
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(),
    axis.line.x = element_line(color = "black", linewidth = 0.4),
    axis.line.y = element_line(color = "black", linewidth = 0.4))+
  scale_x_continuous(limits = c(0,5), expand=(c(0.01, 0.01))) +
  scale_y_continuous(limits = c(0,0.2), expand=c(0.005, 0.005))

plot_subsite

```

Warning: Removed 668 rows containing missing values or values outside the scale range (``geom_ribbon()``).

Warning: Removed 668 rows containing missing values or values outside the scale range (``geom_line()``).



```
# Plot cumulative incidence of cancer death by race and ethnicity
# For ease of plotting for other covariates, use generic plot code after setting
df_age <- sf_age

# Create a data matrix with the correct number of rows:
# A 0 for each group plus the sum of the number of rows for each subgroup.
# Drop "unknown" race.
DATA_ci_age <- matrix(NA, nrow=(4 + sum(df_age$strata[1:4])), ncol=4)
# x
DATA_ci_age[,1] <- c(0, df_age$time[1:df_age$strata[1]],
  0, df_age$time[(1 + df_age$strata[1]):(sum(df_age$strata[1:2]))],
  0, df_age$time[(1 + sum(df_age$strata[1:2])):(sum(df_age$strata[1:3]))],
  0, df_age$time[(1 + sum(df_age$strata[1:3])):(sum(df_age$strata[1:4]))])
# y
DATA_ci_age[,2] <- c(0, df_age$pstate[1:df_age$strata[1],2],
  0, df_age$pstate[(1 + df_age$strata[1]):(sum(df_age$strata[1:2])), 2],
  0, df_age$pstate[(1 + sum(df_age$strata[1:2])):(sum(df_age$strata[1:3])), 2],
  0, df_age$pstate[(1 + sum(df_age$strata[1:3])):(sum(df_age$strata[1:4])), 2])
# 95% CI (lower bound)
DATA_ci_age[,3] <- c(0, df_age$lower[1:df_age$strata[1],2],
```

```

0, df_age$lower[(1 + df_age$strata[1]):(sum(df_age$strata[1:2])), 2],
0, df_age$lower[(1 + sum(df_age$strata[1:2])):(sum(df_age$strata[1:3])), 2],
0, df_age$lower[(1 + sum(df_age$strata[1:3])):(sum(df_age$strata[1:4])), 2])
# 95% CI (upper bound)
DATA_ci_age[,4] <- c(0,df_age$upper[1:df_age$strata[1],2],
0, df_age$upper[(1 + df_age$strata[1]):(sum(df_age$strata[1:2])), 2],
0, df_age$upper[(1 + sum(df_age$strata[1:2])):(sum(df_age$strata[1:3])), 2],
0, df_age$upper[(1 + sum(df_age$strata[1:3])):(sum(df_age$strata[1:4])), 2])
DATA_ci_age <- as.data.frame(DATA_ci_age)
colnames(DATA_ci_age) <- c("x","y","ymin","ymax")
breaks <- which(c(DATA_ci_age$x,0) == 0)[-1]
DATA_ci_age$age <- c(rep("<50", 1+df_age$strata[1]),
rep("50-59", 1+ df_age$strata[2]),
rep("60-69", 1+ df_age$strata[3]),
rep("+70", 1+ df_age$strata[4]))
DATA_ci_age$age <- factor(DATA_ci_age$age)

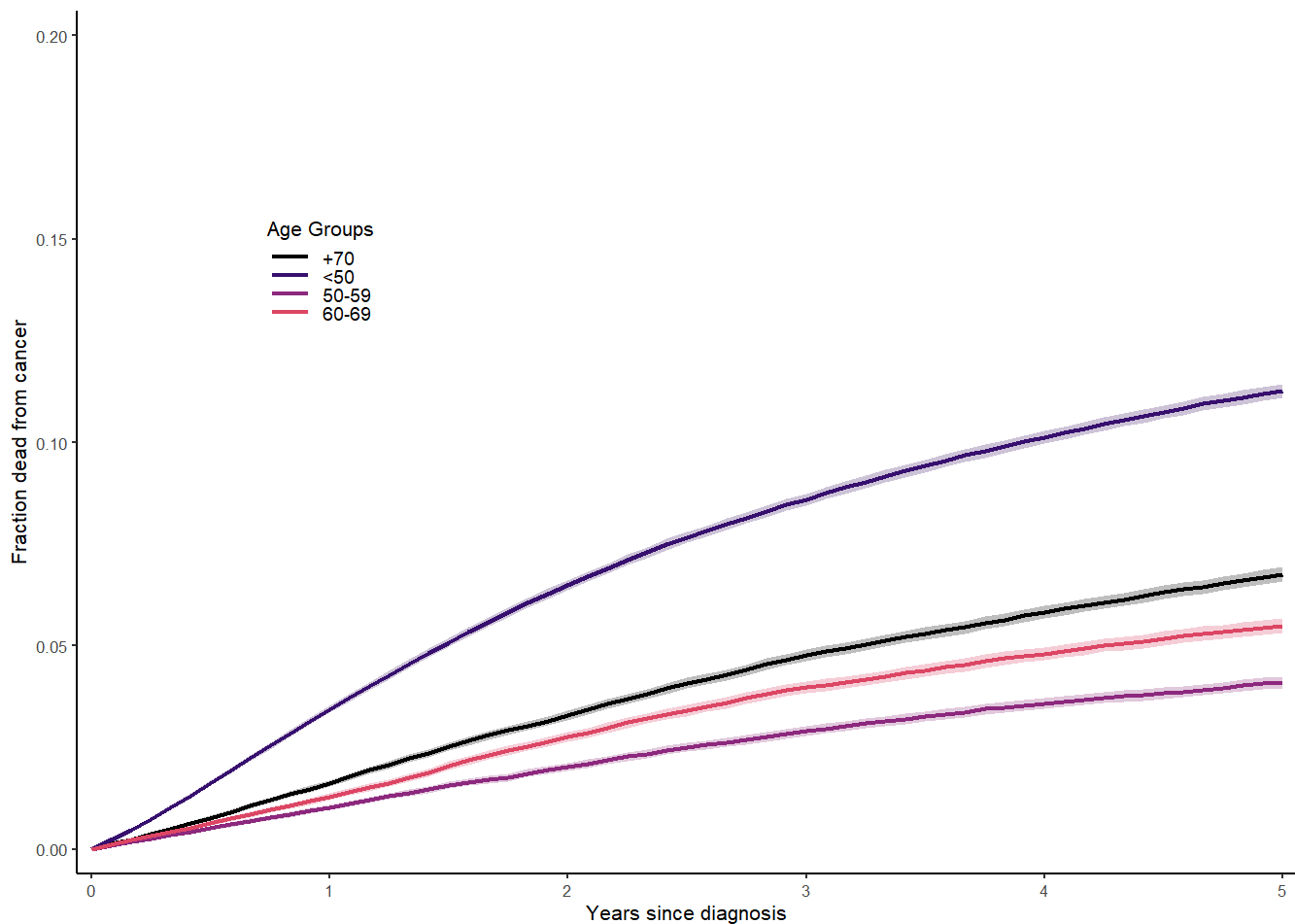
# In this plot, we have two groups, so it gets busy to also show other
# cause deaths. We drop them from this plot.
plot_age <- ggplot(data = DATA_ci_age, aes(x = x, y = y, ymin = ymin, ymax = ymax, color = age)) +
scale_color_manual(values = mycolors(6)[-6], name = "Age Groups") +
geom_ribbon(aes(fill = age, linetype = NA), alpha = 0.25, show.legend = FALSE) +
geom_line(linewidth = 0.75) +
scale_fill_manual(guide="none", values = mycolors(6)[-6]) +
theme_bw(base_size = 8) + labs(x = 'Years since diagnosis', y = 'Fraction dead from cancer') +
theme(axis.title.y = element_text(vjust = 1.5)) +
theme(legend.position = "inside",
legend.position.inside = c(0.2,0.7), legend.key = element_blank(),
legend.background = element_blank(), legend.text = element_text(size=7),
legend.key.height=unit(0.05,"inch")) +
theme(panel.border = element_blank(),
panel.background = element_blank(),
panel.grid.major = element_blank(),
panel.grid.minor = element_blank(),
axis.line.x = element_line(color = "black", linewidth = 0.4),
axis.line.y = element_line(color = "black", linewidth = 0.4))+
scale_x_continuous(limits = c(0, 5), expand = (c(0.01, 0.01)))+
scale_y_continuous(limits = c(0, 0.2), expand = c(0.005, 0.005))

plot_age

```

Warning: Removed 668 rows containing missing values or values outside the scale range (`geom\_ribbon()`).

Warning: Removed 668 rows containing missing values or values outside the scale range (`geom\_line()`).



4/ Cox proportional hazards models

We saw in the above plots that covariates impact the survival. If we assume that the rates are proportional over time, what are the hazard ratios?

We will focus on survival in the first 5 years.

```
# Define a censoring variable at 5 years survival
data$censored.time <- ifelse(data$time > 5, 5, data$time)
data$censored.event <- ifelse(data$time > 5, "Alive", as.character(data$event))

# Because we are considering the instantaneous rate of cancer death,
# we treat other deaths as being censored
data$cancer.death <- (data$censored.event == "Cancer death")

# Generate survival object
cancer.death.surv.5yr <- Surv(data$censored.time, data$cancer.death)

# By sex male
fit.coxph.sex <- coxph(cancer.death.surv.5yr ~ sex,
                      data = data)
summary(fit.coxph.sex)
```

Call:

```
coxph(formula = cancer.death.surv.5yr ~ sex, data = data)
```

n= 416918, number of events= 27689

	coef	exp(coef)	se(coef)	z	Pr(> z)
sexMale	0.49774	1.64500	0.01306	38.12	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
sexMale	1.645	0.6079	1.603	1.688

Concordance= 0.557 (se = 0.001)

Likelihood ratio test= 1538 on 1 df, p=<2e-16

Wald test = 1453 on 1 df, p=<2e-16

Score (logrank) test = 1484 on 1 df, p=<2e-16

```
# The hazard ratio is exp(coef). Is it significant?
```

```
# Stage is the most important predictor of stage, so other variables are
```

```
# vulnerable to confounding. We should adjust for stage.
```

```
fit.coxph.sex_stage <- coxph(cancer.death.surv.5yr ~ sex + stage,
                             data = data)
```

```
summary(fit.coxph.sex_stage)
```

Call:

```
coxph(formula = cancer.death.surv.5yr ~ sex + stage, data = data)
```

n= 416918, number of events= 27689

	coef	exp(coef)	se(coef)	z	Pr(> z)
sexMale	0.40096	1.49325	0.01308	30.66	<2e-16 ***
stageRegional	2.07687	7.97947	0.01394	148.97	<2e-16 ***
stageDistant	3.20209	24.58386	0.01761	181.84	<2e-16 ***
stageUnstaged	0.69945	2.01265	0.02493	28.06	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
sexMale	1.493	0.66968	1.455	1.532
stageRegional	7.979	0.12532	7.764	8.201
stageDistant	24.584	0.04068	23.750	25.447
stageUnstaged	2.013	0.49686	1.917	2.113

Concordance= 0.748 (se = 0.002)

Likelihood ratio test= 33986 on 4 df, p=<2e-16

Wald test = 44341 on 4 df, p=<2e-16

Score (logrank) test = 78629 on 4 df, p=<2e-16

```
fit.coxph.sex_full <- coxph(cancer.death.surv.5yr ~ sex + stage + age + yod + race + event + subsite,
                             data = data)
```

Warning in coxph.fit(X, Y, istrat, offset, init, control, weights = weights, :
Loglik converged before variable 14 ; coefficient may be infinite.

```
summary(fit.coxph.sex_full)
```

Call:

```
coxph(formula = cancer.death.surv.5yr ~ sex + stage + age + yod +
      race + event + subsite, data = data)
```

n= 415909, number of events= 27536
(1009 observations deleted due to missingness)

	coef	exp(coef)	se(coef)
sexMale	1.835e-02	1.019e+00	1.349e-02
stageRegional	5.586e-01	1.748e+00	1.420e-02
stageDistant	1.358e+00	3.889e+00	1.806e-02
stageUnstaged	3.828e-01	1.466e+00	2.509e-02
age<50	-4.221e-01	6.557e-01	2.042e-02
age50-59	-4.010e-01	6.696e-01	1.884e-02
age60-69	-3.290e-01	7.197e-01	1.580e-02
yod2010-2022	3.549e-01	1.426e+00	1.279e-02
raceHispanic (All Races)	9.038e-02	1.095e+00	2.640e-02
raceNon-Hispanic American Indian/Alaska Native	2.347e-02	1.024e+00	1.003e-01
raceNon-Hispanic Asian or Pacific Islander	-3.176e-02	9.687e-01	4.634e-02
raceNon-Hispanic Black	1.643e-01	1.179e+00	3.889e-02
raceNon-Hispanic Unknown Race	-4.740e-02	9.537e-01	1.493e-01
eventCancer death	2.207e+01	3.863e+09	1.154e+02
eventOther death	1.558e-01	1.169e+00	2.373e+02
subsiteTrunk	-5.414e-03	9.946e-01	1.570e-02
subsiteArm	-8.475e-02	9.187e-01	1.743e-02
subsiteLeg	-8.197e-02	9.213e-01	1.820e-02

	z	Pr(> z)
sexMale	1.360	0.173743
stageRegional	39.343	< 2e-16 ***
stageDistant	75.198	< 2e-16 ***
stageUnstaged	15.260	< 2e-16 ***
age<50	-20.666	< 2e-16 ***
age50-59	-21.289	< 2e-16 ***
age60-69	-20.817	< 2e-16 ***
yod2010-2022	27.746	< 2e-16 ***
raceHispanic (All Races)	3.424	0.000617 ***
raceNon-Hispanic American Indian/Alaska Native	0.234	0.815015
raceNon-Hispanic Asian or Pacific Islander	-0.685	0.493139
raceNon-Hispanic Black	4.224	2.40e-05 ***
raceNon-Hispanic Unknown Race	-0.317	0.750891

eventCancer death	0.191 0.848262
eventOther death	0.001 0.999476
subsiteTrunk	-0.345 0.730132
subsiteArm	-4.862 1.16e-06 ***
subsiteLeg	-4.504 6.66e-06 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95
sexMale	1.019e+00	9.818e-01	9.919e-01
stageRegional	1.748e+00	5.720e-01	1.700e+00
stageDistant	3.889e+00	2.571e-01	3.754e+00
stageUnstaged	1.466e+00	6.819e-01	1.396e+00
age<50	6.557e-01	1.525e+00	6.300e-01
age50-59	6.696e-01	1.493e+00	6.454e-01
age60-69	7.197e-01	1.390e+00	6.977e-01
yod2010-2022	1.426e+00	7.013e-01	1.391e+00
raceHispanic (All Races)	1.095e+00	9.136e-01	1.039e+00
raceNon-Hispanic American Indian/Alaska Native	1.024e+00	9.768e-01	8.410e-01
raceNon-Hispanic Asian or Pacific Islander	9.687e-01	1.032e+00	8.846e-01
raceNon-Hispanic Black	1.179e+00	8.485e-01	1.092e+00
raceNon-Hispanic Unknown Race	9.537e-01	1.049e+00	7.118e-01
eventCancer death	3.863e+09	2.588e-10	2.414e-89
eventOther death	1.169e+00	8.557e-01	1.150e-202
subsiteTrunk	9.946e-01	1.005e+00	9.645e-01
subsiteArm	9.187e-01	1.088e+00	8.879e-01
subsiteLeg	9.213e-01	1.085e+00	8.890e-01
	upper .95		
sexMale	1.046e+00		
stageRegional	1.798e+00		
stageDistant	4.029e+00		
stageUnstaged	1.540e+00		
age<50	6.825e-01		
age50-59	6.948e-01		
age60-69	7.423e-01		
yod2010-2022	1.462e+00		
raceHispanic (All Races)	1.153e+00		
raceNon-Hispanic American Indian/Alaska Native	1.246e+00		
raceNon-Hispanic Asian or Pacific Islander	1.061e+00		
raceNon-Hispanic Black	1.272e+00		
raceNon-Hispanic Unknown Race	1.278e+00		
eventCancer death	6.182e+107		
eventOther death	1.187e+202		
subsiteTrunk	1.026e+00		
subsiteArm	9.507e-01		
subsiteLeg	9.548e-01		

Concordance= 0.979 (se = 0)

Likelihood ratio test= 162841 on 18 df, p=<2e-16

Wald test = 7886 on 18 df, $p < 2e-16$
 Score (logrank) test = 450381 on 18 df, $p < 2e-16$

```
# Did the hazard ratio for sex increase or decrease?
# What if we included all the covariates in a multivariable model?
```

```
data_sex_female = data
data_sex_female$sex = relevel(data$sex, ref = "Male")

# By sex female
fit.coxph.sex_female <- coxph(cancer.death.surv.5yr ~ sex,
                              data = data_sex_female)
summary(fit.coxph.sex_female)
```

Call:

```
coxph(formula = cancer.death.surv.5yr ~ sex, data = data_sex_female)
```

n= 416918, number of events= 27689

	coef	exp(coef)	se(coef)	z	Pr(> z)
sexFemale	-0.49774	0.60790	0.01306	-38.12	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
sexFemale	0.6079	1.645	0.5925	0.6237

Concordance= 0.557 (se = 0.001)

Likelihood ratio test= 1538 on 1 df, $p < 2e-16$

Wald test = 1453 on 1 df, $p < 2e-16$

Score (logrank) test = 1484 on 1 df, $p < 2e-16$

```
# Stage is the most important predictor of stage, so other variable are
# vulnerable to confounding. We should adjust for stage.
fit.coxph.sex_female_stage <- coxph(cancer.death.surv.5yr ~ sex + stage,
                                     data = data_sex_female)
summary(fit.coxph.sex_female_stage)
```

Call:

```
coxph(formula = cancer.death.surv.5yr ~ sex + stage, data = data_sex_female)
```

n= 416918, number of events= 27689

	coef	exp(coef)	se(coef)	z	Pr(> z)
sexFemale	-0.40096	0.66968	0.01308	-30.66	<2e-16 ***
stageRegional	2.07687	7.97947	0.01394	148.97	<2e-16 ***
stageDistant	3.20209	24.58386	0.01761	181.84	<2e-16 ***
stageUnstaged	0.69945	2.01265	0.02493	28.06	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
sexFemale	0.6697	1.49325	0.6527	0.6871
stageRegional	7.9795	0.12532	7.7644	8.2005
stageDistant	24.5839	0.04068	23.7499	25.4471
stageUnstaged	2.0127	0.49686	1.9167	2.1134

Concordance= 0.748 (se = 0.002)

Likelihood ratio test= 33986 on 4 df, p=<2e-16

Wald test = 44341 on 4 df, p=<2e-16

Score (logrank) test = 78629 on 4 df, p=<2e-16

```
fit.coxph.sex_female_full <- coxph(cancer.death.surv.5yr ~ sex + stage + age + yod + race + event
                                   data = data_sex_female)
```

Warning in coxph.fit(X, Y, istrat, offset, init, control, weights = weights, :
Loglik converged before variable 14 ; coefficient may be infinite.

```
summary(fit.coxph.sex_female_full)
```

Call:

```
coxph(formula = cancer.death.surv.5yr ~ sex + stage + age + yod +
      race + event + subsite, data = data_sex_female)
```

n= 415909, number of events= 27536
(1009 observations deleted due to missingness)

	coef	exp(coef)	se(coef)
sexFemale	-1.835e-02	9.818e-01	1.349e-02
stageRegional	5.586e-01	1.748e+00	1.420e-02
stageDistant	1.358e+00	3.889e+00	1.806e-02
stageUnstaged	3.828e-01	1.466e+00	2.509e-02
age<50	-4.221e-01	6.557e-01	2.042e-02
age50-59	-4.010e-01	6.696e-01	1.884e-02
age60-69	-3.290e-01	7.197e-01	1.580e-02
yod2010-2022	3.549e-01	1.426e+00	1.279e-02
raceHispanic (All Races)	9.038e-02	1.095e+00	2.640e-02
raceNon-Hispanic American Indian/Alaska Native	2.347e-02	1.024e+00	1.003e-01
raceNon-Hispanic Asian or Pacific Islander	-3.176e-02	9.687e-01	4.634e-02
raceNon-Hispanic Black	1.643e-01	1.179e+00	3.889e-02
raceNon-Hispanic Unknown Race	-4.740e-02	9.537e-01	1.493e-01
eventCancer death	2.207e+01	3.863e+09	1.154e+02
eventOther death	1.558e-01	1.169e+00	2.373e+02
subsiteTrunk	-5.414e-03	9.946e-01	1.570e-02
subsiteArm	-8.475e-02	9.187e-01	1.743e-02
subsiteLeg	-8.197e-02	9.213e-01	1.820e-02

z Pr(>|z|)

sexFemale	-1.360	0.173743	
stageRegional	39.343	< 2e-16	***
stageDistant	75.198	< 2e-16	***
stageUnstaged	15.260	< 2e-16	***
age<50	-20.666	< 2e-16	***
age50-59	-21.289	< 2e-16	***
age60-69	-20.817	< 2e-16	***
yod2010-2022	27.746	< 2e-16	***
raceHispanic (All Races)	3.424	0.000617	***
raceNon-Hispanic American Indian/Alaska Native	0.234	0.815015	
raceNon-Hispanic Asian or Pacific Islander	-0.685	0.493139	
raceNon-Hispanic Black	4.224	2.40e-05	***
raceNon-Hispanic Unknown Race	-0.317	0.750891	
eventCancer death	0.191	0.848266	
eventOther death	0.001	0.999476	
subsiteTrunk	-0.345	0.730132	
subsiteArm	-4.862	1.16e-06	***
subsiteLeg	-4.504	6.66e-06	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95
sexFemale	9.818e-01	1.019e+00	9.562e-01
stageRegional	1.748e+00	5.720e-01	1.700e+00
stageDistant	3.889e+00	2.571e-01	3.754e+00
stageUnstaged	1.466e+00	6.819e-01	1.396e+00
age<50	6.557e-01	1.525e+00	6.300e-01
age50-59	6.696e-01	1.493e+00	6.454e-01
age60-69	7.197e-01	1.390e+00	6.977e-01
yod2010-2022	1.426e+00	7.013e-01	1.391e+00
raceHispanic (All Races)	1.095e+00	9.136e-01	1.039e+00
raceNon-Hispanic American Indian/Alaska Native	1.024e+00	9.768e-01	8.410e-01
raceNon-Hispanic Asian or Pacific Islander	9.687e-01	1.032e+00	8.846e-01
raceNon-Hispanic Black	1.179e+00	8.485e-01	1.092e+00
raceNon-Hispanic Unknown Race	9.537e-01	1.049e+00	7.118e-01
eventCancer death	3.863e+09	2.588e-10	2.397e-89
eventOther death	1.169e+00	8.557e-01	1.146e-202
subsiteTrunk	9.946e-01	1.005e+00	9.645e-01
subsiteArm	9.187e-01	1.088e+00	8.879e-01
subsiteLeg	9.213e-01	1.085e+00	8.890e-01

upper .95

sexFemale	1.008e+00
stageRegional	1.798e+00
stageDistant	4.029e+00
stageUnstaged	1.540e+00
age<50	6.825e-01
age50-59	6.948e-01
age60-69	7.423e-01
yod2010-2022	1.462e+00
raceHispanic (All Races)	1.153e+00
raceNon-Hispanic American Indian/Alaska Native	1.246e+00

raceNon-Hispanic Asian or Pacific Islander	1.061e+00
raceNon-Hispanic Black	1.272e+00
raceNon-Hispanic Unknown Race	1.278e+00
eventCancer death	6.225e+107
eventOther death	1.191e+202
subsiteTrunk	1.026e+00
subsiteArm	9.507e-01
subsiteLeg	9.548e-01

Concordance= 0.979 (se = 0)

Likelihood ratio test= 162841 on 18 df, p=<2e-16

Wald test = 7886 on 18 df, p=<2e-16

Score (logrank) test = 450381 on 18 df, p=<2e-16

```
# Define a censoring variable at 5 years survival
# By race (black)
fit.coxph.race_black <- coxph(cancer.death.surv.5yr ~ race,
                             data = data)
summary(fit.coxph.race_black)
```

Call:

```
coxph(formula = cancer.death.surv.5yr ~ race, data = data)
```

n= 416918, number of events= 27689

	coef	exp(coef)	se(coef)
raceHispanic (All Races)	0.41304	1.51141	0.02591
raceNon-Hispanic American Indian/Alaska Native	0.28546	1.33037	0.10020
raceNon-Hispanic Asian or Pacific Islander	0.52299	1.68706	0.04608
raceNon-Hispanic Black	0.72638	2.06758	0.03833
raceNon-Hispanic Unknown Race	-3.08919	0.04554	0.14921

	z	Pr(> z)
raceHispanic (All Races)	15.942	< 2e-16 ***
raceNon-Hispanic American Indian/Alaska Native	2.849	0.00439 **
raceNon-Hispanic Asian or Pacific Islander	11.349	< 2e-16 ***
raceNon-Hispanic Black	18.952	< 2e-16 ***
raceNon-Hispanic Unknown Race	-20.704	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95
raceHispanic (All Races)	1.51141	0.6616	1.43657
raceNon-Hispanic American Indian/Alaska Native	1.33037	0.7517	1.09315
raceNon-Hispanic Asian or Pacific Islander	1.68706	0.5927	1.54136
raceNon-Hispanic Black	2.06758	0.4837	1.91795
raceNon-Hispanic Unknown Race	0.04554	21.9592	0.03399
	upper .95		
raceHispanic (All Races)	1.59014		
raceNon-Hispanic American Indian/Alaska Native	1.61906		
raceNon-Hispanic Asian or Pacific Islander	1.84653		

raceNon-Hispanic Black	2.22888
raceNon-Hispanic Unknown Race	0.06101

Concordance= 0.536 (se = 0.001)

Likelihood ratio test= 2256 on 5 df, p=<2e-16

Wald test = 1147 on 5 df, p=<2e-16

Score (logrank) test = 1691 on 5 df, p=<2e-16

```
# The hazard ratio is exp(coef). Is is significant?

# Stage is the most important predictor of stage, so other variable are
# vulnerable to confounding. We should adjust for stage.
fit.coxph.race_black_stage <- coxph(cancer.death.surv.5yr ~ race + stage,
                                   data = data)
summary(fit.coxph.race_black_stage)
```

Call:

```
coxph(formula = cancer.death.surv.5yr ~ race + stage, data = data)
```

n= 416918, number of events= 27689

	coef	exp(coef)	se(coef)
raceHispanic (All Races)	0.11721	1.12435	0.02599
raceNon-Hispanic American Indian/Alaska Native	0.16138	1.17513	0.10021
raceNon-Hispanic Asian or Pacific Islander	0.12071	1.12830	0.04618
raceNon-Hispanic Black	0.29428	1.34216	0.03847
raceNon-Hispanic Unknown Race	-2.78183	0.06193	0.14936
stageRegional	2.06089	7.85294	0.01396
stageDistant	3.18799	24.23956	0.01773
stageUnstaged	0.77506	2.17073	0.02496

	z	Pr(> z)
raceHispanic (All Races)	4.509	6.51e-06 ***
raceNon-Hispanic American Indian/Alaska Native	1.610	0.10730
raceNon-Hispanic Asian or Pacific Islander	2.614	0.00896 **
raceNon-Hispanic Black	7.649	2.02e-14 ***
raceNon-Hispanic Unknown Race	-18.625	< 2e-16 ***
stageRegional	147.594	< 2e-16 ***
stageDistant	179.808	< 2e-16 ***
stageUnstaged	31.054	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95
raceHispanic (All Races)	1.12435	0.88940	1.06851
raceNon-Hispanic American Indian/Alaska Native	1.17513	0.85097	0.96558
raceNon-Hispanic Asian or Pacific Islander	1.12830	0.88629	1.03065
raceNon-Hispanic Black	1.34216	0.74507	1.24468
raceNon-Hispanic Unknown Race	0.06193	16.14853	0.04621
stageRegional	7.85294	0.12734	7.64094
stageDistant	24.23956	0.04125	23.41171

stageUnstaged	2.17073	0.46068	2.06709
	upper .95		
raceHispanic (All Races)	1.18312		
raceNon-Hispanic American Indian/Alaska Native	1.43015		
raceNon-Hispanic Asian or Pacific Islander	1.23520		
raceNon-Hispanic Black	1.44728		
raceNon-Hispanic Unknown Race	0.08299		
stageRegional	8.07082		
stageDistant	25.09669		
stageUnstaged	2.27955		

Concordance= 0.736 (se = 0.002)

Likelihood ratio test= 34174 on 8 df, p=<2e-16

Wald test = 43106 on 8 df, p=<2e-16

Score (logrank) test = 78274 on 8 df, p=<2e-16

```
# Stage is the most important predictor of stage, so other variable are
# vulnerable to confounding. We should adjust for stage.
fit.coxph.race_black_full <- coxph(cancer.death.surv.5yr ~ race + stage + age + yod + sex + event
                                   data = data)
```

Warning in coxph.fit(X, Y, istrat, offset, init, control, weights = weights, :
Loglik converged before variable 14 ; coefficient may be infinite.

```
summary(fit.coxph.race_black_full)
```

Call:

```
coxph(formula = cancer.death.surv.5yr ~ race + stage + age +
      yod + sex + event + subsite, data = data)
```

n= 415909, number of events= 27536
(1009 observations deleted due to missingness)

	coef	exp(coef)	se(coef)
raceHispanic (All Races)	9.038e-02	1.095e+00	2.640e-02
raceNon-Hispanic American Indian/Alaska Native	2.347e-02	1.024e+00	1.003e-01
raceNon-Hispanic Asian or Pacific Islander	-3.176e-02	9.687e-01	4.634e-02
raceNon-Hispanic Black	1.643e-01	1.179e+00	3.889e-02
raceNon-Hispanic Unknown Race	-4.740e-02	9.537e-01	1.493e-01
stageRegional	5.586e-01	1.748e+00	1.420e-02
stageDistant	1.358e+00	3.889e+00	1.806e-02
stageUnstaged	3.828e-01	1.466e+00	2.509e-02
age<50	-4.221e-01	6.557e-01	2.042e-02
age50-59	-4.010e-01	6.696e-01	1.884e-02
age60-69	-3.290e-01	7.197e-01	1.580e-02
yod2010-2022	3.549e-01	1.426e+00	1.279e-02
sexMale	1.835e-02	1.019e+00	1.349e-02
eventCancer death	2.207e+01	3.863e+09	1.154e+02

eventOther death	1.558e-01	1.169e+00	2.373e+02
subsiteTrunk	-5.414e-03	9.946e-01	1.570e-02
subsiteArm	-8.475e-02	9.187e-01	1.743e-02
subsiteLeg	-8.197e-02	9.213e-01	1.820e-02

z Pr(>|z|)

raceHispanic (All Races)	3.424	0.000617	***
raceNon-Hispanic American Indian/Alaska Native	0.234	0.815015	
raceNon-Hispanic Asian or Pacific Islander	-0.685	0.493139	
raceNon-Hispanic Black	4.224	2.40e-05	***
raceNon-Hispanic Unknown Race	-0.317	0.750891	
stageRegional	39.343	< 2e-16	***
stageDistant	75.198	< 2e-16	***
stageUnstaged	15.260	< 2e-16	***
age<50	-20.666	< 2e-16	***
age50-59	-21.289	< 2e-16	***
age60-69	-20.817	< 2e-16	***
yod2010-2022	27.746	< 2e-16	***
sexMale	1.360	0.173743	
eventCancer death	0.191	0.848262	
eventOther death	0.001	0.999476	
subsiteTrunk	-0.345	0.730132	
subsiteArm	-4.862	1.16e-06	***
subsiteLeg	-4.504	6.66e-06	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95
raceHispanic (All Races)	1.095e+00	9.136e-01	1.039e+00
raceNon-Hispanic American Indian/Alaska Native	1.024e+00	9.768e-01	8.410e-01
raceNon-Hispanic Asian or Pacific Islander	9.687e-01	1.032e+00	8.846e-01
raceNon-Hispanic Black	1.179e+00	8.485e-01	1.092e+00
raceNon-Hispanic Unknown Race	9.537e-01	1.049e+00	7.118e-01
stageRegional	1.748e+00	5.720e-01	1.700e+00
stageDistant	3.889e+00	2.571e-01	3.754e+00
stageUnstaged	1.466e+00	6.819e-01	1.396e+00
age<50	6.557e-01	1.525e+00	6.300e-01
age50-59	6.696e-01	1.493e+00	6.454e-01
age60-69	7.197e-01	1.390e+00	6.977e-01
yod2010-2022	1.426e+00	7.013e-01	1.391e+00
sexMale	1.019e+00	9.818e-01	9.919e-01
eventCancer death	3.863e+09	2.588e-10	2.414e-89
eventOther death	1.169e+00	8.557e-01	1.150e-202
subsiteTrunk	9.946e-01	1.005e+00	9.645e-01
subsiteArm	9.187e-01	1.088e+00	8.879e-01
subsiteLeg	9.213e-01	1.085e+00	8.890e-01

upper .95

raceHispanic (All Races)	1.153e+00
raceNon-Hispanic American Indian/Alaska Native	1.246e+00
raceNon-Hispanic Asian or Pacific Islander	1.061e+00
raceNon-Hispanic Black	1.272e+00
raceNon-Hispanic Unknown Race	1.278e+00

stageRegional	1.798e+00
stageDistant	4.029e+00
stageUnstaged	1.540e+00
age<50	6.825e-01
age50-59	6.948e-01
age60-69	7.423e-01
yod2010-2022	1.462e+00
sexMale	1.046e+00
eventCancer death	6.182e+107
eventOther death	1.187e+202
subsiteTrunk	1.026e+00
subsiteArm	9.507e-01
subsiteLeg	9.548e-01

Concordance= 0.979 (se = 0)

Likelihood ratio test= 162841 on 18 df, p=<2e-16

Wald test = 7886 on 18 df, p=<2e-16

Score (logrank) test = 450381 on 18 df, p=<2e-16

```
# Did the hazard ratio for sex increase or decrease?
# What if we included all the covariates in a multivariable model?
```

```
data_race_white = data
data_race_white$race = relevel(data$race, ref = "Non-Hispanic Black")

# Define a censoring variable at 5 years survival
# By race white
fit.coxph.race_white <- coxph(cancer.death.surv.5yr ~ race,
                             data = data_race_white)
summary(fit.coxph.race_white)
```

Call:

```
coxph(formula = cancer.death.surv.5yr ~ race, data = data_race_white)
```

n= 416918, number of events= 27689

	coef	exp(coef)	se(coef)
raceNon-Hispanic White	-0.72638	0.48366	0.03833
raceHispanic (All Races)	-0.31334	0.73100	0.04538
raceNon-Hispanic American Indian/Alaska Native	-0.44092	0.64344	0.10690
raceNon-Hispanic Asian or Pacific Islander	-0.20339	0.81596	0.05926
raceNon-Hispanic Unknown Race	-3.81556	0.02203	0.15379

	z	Pr(> z)
raceNon-Hispanic White	-18.952	< 2e-16 ***
raceHispanic (All Races)	-6.905	5.04e-12 ***
raceNon-Hispanic American Indian/Alaska Native	-4.124	3.72e-05 ***
raceNon-Hispanic Asian or Pacific Islander	-3.432	0.000599 ***
raceNon-Hispanic Unknown Race	-24.811	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95
raceNon-Hispanic White	0.48366	2.068	0.44866
raceHispanic (All Races)	0.73100	1.368	0.66879
raceNon-Hispanic American Indian/Alaska Native	0.64344	1.554	0.52181
raceNon-Hispanic Asian or Pacific Islander	0.81596	1.226	0.72648
raceNon-Hispanic Unknown Race	0.02203	45.402	0.01629
	upper .95		
raceNon-Hispanic White	0.52139		
raceHispanic (All Races)	0.79900		
raceNon-Hispanic American Indian/Alaska Native	0.79343		
raceNon-Hispanic Asian or Pacific Islander	0.91646		
raceNon-Hispanic Unknown Race	0.02977		

Concordance= 0.536 (se = 0.001)

Likelihood ratio test= 2256 on 5 df, p=<2e-16

Wald test = 1147 on 5 df, p=<2e-16

Score (logrank) test = 1691 on 5 df, p=<2e-16

```
# The hazard ratio is exp(coef). Is is significant?

# Stage is the most important predictor of stage, so other variable are
# vulnerable to confounding. We should adjust for stage.
fit.coxph.race_white_stage <- coxph(cancer.death.surv.5yr ~ race + stage,
                                   data = data_race_white)
summary(fit.coxph.race_white_stage)
```

Call:

```
coxph(formula = cancer.death.surv.5yr ~ race + stage, data = data_race_white)
```

n= 416918, number of events= 27689

	coef	exp(coef)	se(coef)
raceNon-Hispanic White	-0.29428	0.74507	0.03847
raceHispanic (All Races)	-0.17707	0.83772	0.04541
raceNon-Hispanic American Indian/Alaska Native	-0.13290	0.87555	0.10694
raceNon-Hispanic Asian or Pacific Islander	-0.17357	0.84066	0.05926
raceNon-Hispanic Unknown Race	-3.07611	0.04614	0.15399
stageRegional	2.06089	7.85294	0.01396
stageDistant	3.18799	24.23956	0.01773
stageUnstaged	0.77506	2.17073	0.02496
	z	Pr(> z)	
raceNon-Hispanic White	-7.649	2.02e-14	***
raceHispanic (All Races)	-3.900	9.63e-05	***
raceNon-Hispanic American Indian/Alaska Native	-1.243	0.2140	
raceNon-Hispanic Asian or Pacific Islander	-2.929	0.0034	**
raceNon-Hispanic Unknown Race	-19.976	< 2e-16	***
stageRegional	147.594	< 2e-16	***
stageDistant	179.808	< 2e-16	***

stageUnstaged

31.054 < 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95
raceNon-Hispanic White	0.74507	1.34216	0.69095
raceHispanic (All Races)	0.83772	1.19372	0.76639
raceNon-Hispanic American Indian/Alaska Native	0.87555	1.14214	0.70999
raceNon-Hispanic Asian or Pacific Islander	0.84066	1.18954	0.74847
raceNon-Hispanic Unknown Race	0.04614	21.67395	0.03412
stageRegional	7.85294	0.12734	7.64094
stageDistant	24.23956	0.04125	23.41171
stageUnstaged	2.17073	0.46068	2.06709
	upper .95		
raceNon-Hispanic White	0.80342		
raceHispanic (All Races)	0.91569		
raceNon-Hispanic American Indian/Alaska Native	1.07972		
raceNon-Hispanic Asian or Pacific Islander	0.94420		
raceNon-Hispanic Unknown Race	0.06239		
stageRegional	8.07082		
stageDistant	25.09669		
stageUnstaged	2.27955		

Concordance= 0.736 (se = 0.002)

Likelihood ratio test= 34174 on 8 df, p=<2e-16

Wald test = 43106 on 8 df, p=<2e-16

Score (logrank) test = 78274 on 8 df, p=<2e-16

```
# Stage is the most important predictor of stage, so other variable are
# vulnerable to confounding. We should adjust for stage.
fit.coxph.race_white_full <- coxph(cancer.death.surv.5yr ~ race + stage + age + yod + sex + event
                                data = data_race_white)
```

Warning in coxph.fit(X, Y, istrat, offset, init, control, weights = weights, :
Loglik converged before variable 14 ; coefficient may be infinite.

```
summary(fit.coxph.race_white_full)
```

Call:

```
coxph(formula = cancer.death.surv.5yr ~ race + stage + age +
      yod + sex + event + subsite, data = data_race_white)
```

n= 415909, number of events= 27536
(1009 observations deleted due to missingness)

	coef	exp(coef)	se(coef)
raceNon-Hispanic White	-1.643e-01	8.485e-01	3.889e-02
raceHispanic (All Races)	-7.388e-02	9.288e-01	4.555e-02

raceNon-Hispanic American Indian/Alaska Native	-1.408e-01	8.687e-01	1.070e-01
raceNon-Hispanic Asian or Pacific Islander	-1.960e-01	8.220e-01	5.940e-02
raceNon-Hispanic Unknown Race	-2.117e-01	8.092e-01	1.540e-01
stageRegional	5.586e-01	1.748e+00	1.420e-02
stageDistant	1.358e+00	3.889e+00	1.806e-02
stageUnstaged	3.828e-01	1.466e+00	2.509e-02
age<50	-4.221e-01	6.557e-01	2.042e-02
age50-59	-4.010e-01	6.696e-01	1.884e-02
age60-69	-3.290e-01	7.197e-01	1.580e-02
yod2010-2022	3.549e-01	1.426e+00	1.279e-02
sexMale	1.835e-02	1.019e+00	1.349e-02
eventCancer death	2.207e+01	3.863e+09	1.154e+02
eventOther death	1.558e-01	1.169e+00	2.373e+02
subsiteTrunk	-5.414e-03	9.946e-01	1.570e-02
subsiteArm	-8.475e-02	9.187e-01	1.743e-02
subsiteLeg	-8.197e-02	9.213e-01	1.820e-02

z Pr(>|z|)

raceNon-Hispanic White	-4.224	2.40e-05	***
raceHispanic (All Races)	-1.622	0.104804	
raceNon-Hispanic American Indian/Alaska Native	-1.315	0.188443	
raceNon-Hispanic Asian or Pacific Islander	-3.300	0.000968	***
raceNon-Hispanic Unknown Race	-1.374	0.169346	
stageRegional	39.343	< 2e-16	***
stageDistant	75.198	< 2e-16	***
stageUnstaged	15.260	< 2e-16	***
age<50	-20.666	< 2e-16	***
age50-59	-21.289	< 2e-16	***
age60-69	-20.817	< 2e-16	***
yod2010-2022	27.746	< 2e-16	***
sexMale	1.360	0.173743	
eventCancer death	0.191	0.848273	
eventOther death	0.001	0.999476	
subsiteTrunk	-0.345	0.730132	
subsiteArm	-4.862	1.16e-06	***
subsiteLeg	-4.504	6.66e-06	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95
raceNon-Hispanic White	8.485e-01	1.179e+00	7.863e-01
raceHispanic (All Races)	9.288e-01	1.077e+00	8.495e-01
raceNon-Hispanic American Indian/Alaska Native	8.687e-01	1.151e+00	7.043e-01
raceNon-Hispanic Asian or Pacific Islander	8.220e-01	1.217e+00	7.317e-01
raceNon-Hispanic Unknown Race	8.092e-01	1.236e+00	5.984e-01
stageRegional	1.748e+00	5.720e-01	1.700e+00
stageDistant	3.889e+00	2.571e-01	3.754e+00
stageUnstaged	1.466e+00	6.819e-01	1.396e+00
age<50	6.557e-01	1.525e+00	6.300e-01
age50-59	6.696e-01	1.493e+00	6.454e-01
age60-69	7.197e-01	1.390e+00	6.977e-01
yod2010-2022	1.426e+00	7.013e-01	1.391e+00

sexMale	1.019e+00	9.818e-01	9.919e-01
eventCancer death	3.863e+09	2.588e-10	2.375e-89
eventOther death	1.169e+00	8.557e-01	1.141e-202
subsiteTrunk	9.946e-01	1.005e+00	9.645e-01
subsiteArm	9.187e-01	1.088e+00	8.879e-01
subsiteLeg	9.213e-01	1.085e+00	8.890e-01
	upper .95		
raceNon-Hispanic White	9.157e-01		
raceHispanic (All Races)	1.016e+00		
raceNon-Hispanic American Indian/Alaska Native	1.071e+00		
raceNon-Hispanic Asian or Pacific Islander	9.235e-01		
raceNon-Hispanic Unknown Race	1.094e+00		
stageRegional	1.798e+00		
stageDistant	4.029e+00		
stageUnstaged	1.540e+00		
age<50	6.825e-01		
age50-59	6.948e-01		
age60-69	7.423e-01		
yod2010-2022	1.462e+00		
sexMale	1.046e+00		
eventCancer death	6.285e+107		
eventOther death	1.197e+202		
subsiteTrunk	1.026e+00		
subsiteArm	9.507e-01		
subsiteLeg	9.548e-01		

Concordance= 0.979 (se = 0)

Likelihood ratio test= 162841 on 18 df, p=<2e-16

Wald test = 7886 on 18 df, p=<2e-16

Score (logrank) test = 450381 on 18 df, p=<2e-16

```
# Did the hazard ratio for sex increase or decrease?
# What if we included all the covariates in a multivariable model?
```

```
# Define a censoring variable at 5 years survival
# By subsite
fit.coxph.subsite <- coxph(cancer.death.surv.5yr ~ subsite,
                           data = data)
summary(fit.coxph.subsite)
```

Call:

```
coxph(formula = cancer.death.surv.5yr ~ subsite, data = data)
```

n= 415909, number of events= 27536

(1009 observations deleted due to missingness)

	coef	exp(coef)	se(coef)	z	Pr(> z)
subsiteTrunk	-0.48210	0.61749	0.01535	-31.40	<2e-16 ***
subsiteArm	-0.65129	0.52137	0.01733	-37.58	<2e-16 ***

```
subsiteLeg    -0.29678    0.74321    0.01738 -17.08    <2e-16 ***
```

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

	exp(coef)	exp(-coef)	lower .95	upper .95
subsiteTrunk	0.6175	1.619	0.5992	0.6364
subsiteArm	0.5214	1.918	0.5040	0.5394
subsiteLeg	0.7432	1.346	0.7183	0.7690

```
Concordance= 0.567 (se = 0.002 )
```

```
Likelihood ratio test= 1680  on 3 df,  p=<2e-16
```

```
Wald test              = 1717  on 3 df,  p=<2e-16
```

```
Score (logrank) test = 1758  on 3 df,  p=<2e-16
```

```
fit.coxph.subsite_stage <- coxph(cancer.death.surv.5yr ~ subsite + stage,
                                   data = data)
summary(fit.coxph.subsite_stage)
```

Call:

```
coxph(formula = cancer.death.surv.5yr ~ subsite + stage, data = data)
```

```
n= 415909, number of events= 27536
```

```
(1009 observations deleted due to missingness)
```

	coef	exp(coef)	se(coef)	z	Pr(> z)
subsiteTrunk	-0.29545	0.74419	0.01542	-19.16	<2e-16 ***
subsiteArm	-0.41841	0.65809	0.01744	-24.00	<2e-16 ***
subsiteLeg	-0.24136	0.78556	0.01741	-13.87	<2e-16 ***
stageRegional	2.07430	7.95895	0.01402	147.94	<2e-16 ***
stageDistant	3.18260	24.10938	0.01775	179.26	<2e-16 ***
stageUnstaged	0.68300	1.97981	0.02504	27.28	<2e-16 ***

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

	exp(coef)	exp(-coef)	lower .95	upper .95
subsiteTrunk	0.7442	1.34373	0.7220	0.7670
subsiteArm	0.6581	1.51955	0.6360	0.6810
subsiteLeg	0.7856	1.27298	0.7592	0.8128
stageRegional	7.9590	0.12564	7.7432	8.1807
stageDistant	24.1094	0.04148	23.2849	24.9631
stageUnstaged	1.9798	0.50510	1.8850	2.0794

```
Concordance= 0.752 (se = 0.002 )
```

```
Likelihood ratio test= 33562  on 6 df,  p=<2e-16
```

```
Wald test              = 44062  on 6 df,  p=<2e-16
```

```
Score (logrank) test = 78345  on 6 df,  p=<2e-16
```

```
fit.coxph.subsite_full <- coxph(cancer.death.surv.5yr ~ subsite + stage + age + yod + sex + event
```

```
data = data)
```

Warning in coxph.fit(X, Y, istrat, offset, init, control, weights = weights, :
Loglik converged before variable 12 ; coefficient may be infinite.

```
summary(fit.coxph.subsite_full)
```

Call:

```
coxph(formula = cancer.death.surv.5yr ~ subsite + stage + age +  
      yod + sex + event + race, data = data)
```

n= 415909, number of events= 27536
(1009 observations deleted due to missingness)

	coef	exp(coef)	se(coef)
subsiteTrunk	-5.414e-03	9.946e-01	1.570e-02
subsiteArm	-8.475e-02	9.187e-01	1.743e-02
subsiteLeg	-8.197e-02	9.213e-01	1.820e-02
stageRegional	5.586e-01	1.748e+00	1.420e-02
stageDistant	1.358e+00	3.889e+00	1.806e-02
stageUnstaged	3.828e-01	1.466e+00	2.509e-02
age<50	-4.221e-01	6.557e-01	2.042e-02
age50-59	-4.010e-01	6.696e-01	1.884e-02
age60-69	-3.290e-01	7.197e-01	1.580e-02
yod2010-2022	3.549e-01	1.426e+00	1.279e-02
sexMale	1.835e-02	1.019e+00	1.349e-02
eventCancer death	2.207e+01	3.863e+09	1.154e+02
eventOther death	1.558e-01	1.169e+00	2.373e+02
raceHispanic (All Races)	9.038e-02	1.095e+00	2.640e-02
raceNon-Hispanic American Indian/Alaska Native	2.347e-02	1.024e+00	1.003e-01
raceNon-Hispanic Asian or Pacific Islander	-3.176e-02	9.687e-01	4.634e-02
raceNon-Hispanic Black	1.643e-01	1.179e+00	3.889e-02
raceNon-Hispanic Unknown Race	-4.740e-02	9.537e-01	1.493e-01

z Pr(>|z|)

subsiteTrunk	-0.345	0.730132	
subsiteArm	-4.862	1.16e-06	***
subsiteLeg	-4.504	6.66e-06	***
stageRegional	39.343	< 2e-16	***
stageDistant	75.198	< 2e-16	***
stageUnstaged	15.260	< 2e-16	***
age<50	-20.666	< 2e-16	***
age50-59	-21.289	< 2e-16	***
age60-69	-20.817	< 2e-16	***
yod2010-2022	27.746	< 2e-16	***
sexMale	1.360	0.173743	
eventCancer death	0.191	0.848262	
eventOther death	0.001	0.999476	
raceHispanic (All Races)	3.424	0.000617	***

raceNon-Hispanic American Indian/Alaska Native	0.234	0.815015
raceNon-Hispanic Asian or Pacific Islander	-0.685	0.493139
raceNon-Hispanic Black	4.224	2.40e-05 ***
raceNon-Hispanic Unknown Race	-0.317	0.750891

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95
subsiteTrunk	9.946e-01	1.005e+00	9.645e-01
subsiteArm	9.187e-01	1.088e+00	8.879e-01
subsiteLeg	9.213e-01	1.085e+00	8.890e-01
stageRegional	1.748e+00	5.720e-01	1.700e+00
stageDistant	3.889e+00	2.571e-01	3.754e+00
stageUnstaged	1.466e+00	6.819e-01	1.396e+00
age<50	6.557e-01	1.525e+00	6.300e-01
age50-59	6.696e-01	1.493e+00	6.454e-01
age60-69	7.197e-01	1.390e+00	6.977e-01
yod2010-2022	1.426e+00	7.013e-01	1.391e+00
sexMale	1.019e+00	9.818e-01	9.919e-01
eventCancer death	3.863e+09	2.588e-10	2.414e-89
eventOther death	1.169e+00	8.557e-01	1.150e-202
raceHispanic (All Races)	1.095e+00	9.136e-01	1.039e+00
raceNon-Hispanic American Indian/Alaska Native	1.024e+00	9.768e-01	8.410e-01
raceNon-Hispanic Asian or Pacific Islander	9.687e-01	1.032e+00	8.846e-01
raceNon-Hispanic Black	1.179e+00	8.485e-01	1.092e+00
raceNon-Hispanic Unknown Race	9.537e-01	1.049e+00	7.118e-01

upper .95

subsiteTrunk	1.026e+00
subsiteArm	9.507e-01
subsiteLeg	9.548e-01
stageRegional	1.798e+00
stageDistant	4.029e+00
stageUnstaged	1.540e+00
age<50	6.825e-01
age50-59	6.948e-01
age60-69	7.423e-01
yod2010-2022	1.462e+00
sexMale	1.046e+00
eventCancer death	6.182e+107
eventOther death	1.187e+202
raceHispanic (All Races)	1.153e+00
raceNon-Hispanic American Indian/Alaska Native	1.246e+00
raceNon-Hispanic Asian or Pacific Islander	1.061e+00
raceNon-Hispanic Black	1.272e+00
raceNon-Hispanic Unknown Race	1.278e+00

Concordance= 0.979 (se = 0)

Likelihood ratio test= 162841 on 18 df, p=<2e-16

Wald test = 7886 on 18 df, p=<2e-16

Score (logrank) test = 450381 on 18 df, p=<2e-16

```
# Define a censoring variable at 5 years survival
# By age
fit.coxph.age <- coxph(cancer.death.surv.5yr ~ age,
                      data = data)
summary(fit.coxph.age)
```

Call:

```
coxph(formula = cancer.death.surv.5yr ~ age, data = data)
```

n= 416918, number of events= 27689

	coef	exp(coef)	se(coef)	z	Pr(> z)
age<50	-1.18514	0.30570	0.01981	-59.81	<2e-16 ***
age50-59	-0.87930	0.41507	0.01848	-47.58	<2e-16 ***
age60-69	-0.65710	0.51835	0.01561	-42.09	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
age<50	0.3057	3.271	0.2941	0.3178
age50-59	0.4151	2.409	0.4003	0.4304
age60-69	0.5184	1.929	0.5027	0.5345

Concordance= 0.624 (se = 0.002)

Likelihood ratio test= 5820 on 3 df, p=<2e-16

Wald test = 5520 on 3 df, p=<2e-16

Score (logrank) test = 5962 on 3 df, p=<2e-16

```
fit.coxph.age_stage <- coxph(cancer.death.surv.5yr ~ age + stage,
                             data = data)
summary(fit.coxph.age_stage)
```

Call:

```
coxph(formula = cancer.death.surv.5yr ~ age + stage, data = data)
```

n= 416918, number of events= 27689

	coef	exp(coef)	se(coef)	z	Pr(> z)
age<50	-1.11101	0.32923	0.01985	-55.98	<2e-16 ***
age50-59	-0.83230	0.43505	0.01850	-45.00	<2e-16 ***
age60-69	-0.61754	0.53927	0.01562	-39.54	<2e-16 ***
stageRegional	2.08241	8.02381	0.01394	149.44	<2e-16 ***
stageDistant	3.17271	23.87204	0.01761	180.19	<2e-16 ***
stageUnstaged	0.64147	1.89928	0.02494	25.72	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
age<50	0.3292	3.03742	0.3167	0.3423

age50-59	0.4350	2.29860	0.4196	0.4511
age60-69	0.5393	1.85437	0.5230	0.5560
stageRegional	8.0238	0.12463	7.8076	8.2460
stageDistant	23.8720	0.04189	23.0623	24.7102
stageUnstaged	1.8993	0.52652	1.8087	1.9944

Concordance= 0.791 (se = 0.001)

Likelihood ratio test= 38114 on 6 df, p=<2e-16

Wald test = 48172 on 6 df, p=<2e-16

Score (logrank) test = 82915 on 6 df, p=<2e-16

```
fit.coxph.age_full <- coxph(cancer.death.surv.5yr ~ age + stage + subsite + yod + sex + event + race,
                             data = data)
```

Warning in coxph.fit(X, Y, istrat, offset, init, control, weights = weights, :
Loglik converged before variable 12 ; coefficient may be infinite.

```
summary(fit.coxph.age_full)
```

Call:

```
coxph(formula = cancer.death.surv.5yr ~ age + stage + subsite +
      yod + sex + event + race, data = data)
```

n= 415909, number of events= 27536
(1009 observations deleted due to missingness)

	coef	exp(coef)	se(coef)
age<50	-4.221e-01	6.557e-01	2.042e-02
age50-59	-4.010e-01	6.696e-01	1.884e-02
age60-69	-3.290e-01	7.197e-01	1.580e-02
stageRegional	5.586e-01	1.748e+00	1.420e-02
stageDistant	1.358e+00	3.889e+00	1.806e-02
stageUnstaged	3.828e-01	1.466e+00	2.509e-02
subsiteTrunk	-5.414e-03	9.946e-01	1.570e-02
subsiteArm	-8.475e-02	9.187e-01	1.743e-02
subsiteLeg	-8.197e-02	9.213e-01	1.820e-02
yod2010-2022	3.549e-01	1.426e+00	1.279e-02
sexMale	1.835e-02	1.019e+00	1.349e-02
eventCancer death	2.207e+01	3.863e+09	1.154e+02
eventOther death	1.558e-01	1.169e+00	2.373e+02
raceHispanic (All Races)	9.038e-02	1.095e+00	2.640e-02
raceNon-Hispanic American Indian/Alaska Native	2.347e-02	1.024e+00	1.003e-01
raceNon-Hispanic Asian or Pacific Islander	-3.176e-02	9.687e-01	4.634e-02
raceNon-Hispanic Black	1.643e-01	1.179e+00	3.889e-02
raceNon-Hispanic Unknown Race	-4.740e-02	9.537e-01	1.493e-01
	z	Pr(> z)	
age<50	-20.666	< 2e-16 ***	
age50-59	-21.289	< 2e-16 ***	

age60-69	-20.817	< 2e-16	***
stageRegional	39.343	< 2e-16	***
stageDistant	75.198	< 2e-16	***
stageUnstaged	15.260	< 2e-16	***
subsiteTrunk	-0.345	0.730132	
subsiteArm	-4.862	1.16e-06	***
subsiteLeg	-4.504	6.66e-06	***
yod2010-2022	27.746	< 2e-16	***
sexMale	1.360	0.173743	
eventCancer death	0.191	0.848262	
eventOther death	0.001	0.999476	
raceHispanic (All Races)	3.424	0.000617	***
raceNon-Hispanic American Indian/Alaska Native	0.234	0.815015	
raceNon-Hispanic Asian or Pacific Islander	-0.685	0.493139	
raceNon-Hispanic Black	4.224	2.40e-05	***
raceNon-Hispanic Unknown Race	-0.317	0.750891	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95
age<50	6.557e-01	1.525e+00	6.300e-01
age50-59	6.696e-01	1.493e+00	6.454e-01
age60-69	7.197e-01	1.390e+00	6.977e-01
stageRegional	1.748e+00	5.720e-01	1.700e+00
stageDistant	3.889e+00	2.571e-01	3.754e+00
stageUnstaged	1.466e+00	6.819e-01	1.396e+00
subsiteTrunk	9.946e-01	1.005e+00	9.645e-01
subsiteArm	9.187e-01	1.088e+00	8.879e-01
subsiteLeg	9.213e-01	1.085e+00	8.890e-01
yod2010-2022	1.426e+00	7.013e-01	1.391e+00
sexMale	1.019e+00	9.818e-01	9.919e-01
eventCancer death	3.863e+09	2.588e-10	2.414e-89
eventOther death	1.169e+00	8.557e-01	1.150e-202
raceHispanic (All Races)	1.095e+00	9.136e-01	1.039e+00
raceNon-Hispanic American Indian/Alaska Native	1.024e+00	9.768e-01	8.410e-01
raceNon-Hispanic Asian or Pacific Islander	9.687e-01	1.032e+00	8.846e-01
raceNon-Hispanic Black	1.179e+00	8.485e-01	1.092e+00
raceNon-Hispanic Unknown Race	9.537e-01	1.049e+00	7.118e-01
	upper .95		
age<50	6.825e-01		
age50-59	6.948e-01		
age60-69	7.423e-01		
stageRegional	1.798e+00		
stageDistant	4.029e+00		
stageUnstaged	1.540e+00		
subsiteTrunk	1.026e+00		
subsiteArm	9.507e-01		
subsiteLeg	9.548e-01		
yod2010-2022	1.462e+00		
sexMale	1.046e+00		
eventCancer death	6.182e+107		

eventOther death	1.187e+202
raceHispanic (All Races)	1.153e+00
raceNon-Hispanic American Indian/Alaska Native	1.246e+00
raceNon-Hispanic Asian or Pacific Islander	1.061e+00
raceNon-Hispanic Black	1.272e+00
raceNon-Hispanic Unknown Race	1.278e+00

Concordance= 0.979 (se = 0)

Likelihood ratio test= 162841 on 18 df, p=<2e-16

Wald test = 7886 on 18 df, p=<2e-16

Score (logrank) test = 450381 on 18 df, p=<2e-16

```
data_age_70 <- data
data_age_70$age <- relevel(data$age, ref = "<50")

# Define a censoring variable at 5 years survival
# By age greater 70
fit.coxph.age_70 <- coxph(cancer.death.surv.5yr ~ age,
                          data = data_age_70)
summary(fit.coxph.age_70)
```

Call:

```
coxph(formula = cancer.death.surv.5yr ~ age, data = data_age_70)
```

n= 416918, number of events= 27689

	coef	exp(coef)	se(coef)	z	Pr(> z)
age70+	1.18514	3.27113	0.01981	59.81	<2e-16 ***
age50-59	0.30583	1.35775	0.02455	12.46	<2e-16 ***
age60-69	0.52804	1.69560	0.02248	23.49	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
age70+	3.271	0.3057	3.147	3.401
age50-59	1.358	0.7365	1.294	1.425
age60-69	1.696	0.5898	1.623	1.772

Concordance= 0.624 (se = 0.002)

Likelihood ratio test= 5820 on 3 df, p=<2e-16

Wald test = 5520 on 3 df, p=<2e-16

Score (logrank) test = 5962 on 3 df, p=<2e-16

```
fit.coxph.age_70_stage <- coxph(cancer.death.surv.5yr ~ age + stage,
                              data = data_age_70)
summary(fit.coxph.age_70_stage)
```

Call:

```
coxph(formula = cancer.death.surv.5yr ~ age + stage, data = data_age_70)
```

n= 416918, number of events= 27689

	coef	exp(coef)	se(coef)	z	Pr(> z)
age70+	1.11101	3.03742	0.01985	55.98	<2e-16 ***
age50-59	0.27871	1.32142	0.02456	11.35	<2e-16 ***
age60-69	0.49347	1.63798	0.02249	21.94	<2e-16 ***
stageRegional	2.08241	8.02381	0.01394	149.44	<2e-16 ***
stageDistant	3.17271	23.87204	0.01761	180.19	<2e-16 ***
stageUnstaged	0.64147	1.89928	0.02494	25.72	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
age70+	3.037	0.32923	2.922	3.158
age50-59	1.321	0.75676	1.259	1.387
age60-69	1.638	0.61051	1.567	1.712
stageRegional	8.024	0.12463	7.808	8.246
stageDistant	23.872	0.04189	23.062	24.710
stageUnstaged	1.899	0.52652	1.809	1.994

Concordance= 0.791 (se = 0.001)

Likelihood ratio test= 38114 on 6 df, p=<2e-16

Wald test = 48172 on 6 df, p=<2e-16

Score (logrank) test = 82915 on 6 df, p=<2e-16

```
fit.coxph.age_70_full <- coxph(cancer.death.surv.5yr ~ age + stage + subsite + yod + sex + event +
                                data = data_age_70)
```

Warning in coxph.fit(X, Y, istrat, offset, init, control, weights = weights, :
Loglik converged before variable 12 ; coefficient may be infinite.

```
summary(fit.coxph.age_70_full)
```

Call:

```
coxph(formula = cancer.death.surv.5yr ~ age + stage + subsite +
      yod + sex + event + race, data = data_age_70)
```

n= 415909, number of events= 27536

(1009 observations deleted due to missingness)

	coef	exp(coef)	se(coef)
age70+	4.221e-01	1.525e+00	2.042e-02
age50-59	2.108e-02	1.021e+00	2.469e-02
age60-69	9.309e-02	1.098e+00	2.277e-02
stageRegional	5.586e-01	1.748e+00	1.420e-02
stageDistant	1.358e+00	3.889e+00	1.806e-02
stageUnstaged	3.828e-01	1.466e+00	2.509e-02

subsiteTrunk	-5.414e-03	9.946e-01	1.570e-02
subsiteArm	-8.475e-02	9.187e-01	1.743e-02
subsiteLeg	-8.197e-02	9.213e-01	1.820e-02
yod2010-2022	3.549e-01	1.426e+00	1.279e-02
sexMale	1.835e-02	1.019e+00	1.349e-02
eventCancer death	2.207e+01	3.863e+09	1.154e+02
eventOther death	1.558e-01	1.169e+00	2.373e+02
raceHispanic (All Races)	9.038e-02	1.095e+00	2.640e-02
raceNon-Hispanic American Indian/Alaska Native	2.347e-02	1.024e+00	1.003e-01
raceNon-Hispanic Asian or Pacific Islander	-3.176e-02	9.687e-01	4.634e-02
raceNon-Hispanic Black	1.643e-01	1.179e+00	3.889e-02
raceNon-Hispanic Unknown Race	-4.740e-02	9.537e-01	1.493e-01

z Pr(>|z|)

age70+	20.666	< 2e-16	***
age50-59	0.854	0.393293	
age60-69	4.088	4.35e-05	***
stageRegional	39.343	< 2e-16	***
stageDistant	75.198	< 2e-16	***
stageUnstaged	15.260	< 2e-16	***
subsiteTrunk	-0.345	0.730132	
subsiteArm	-4.862	1.16e-06	***
subsiteLeg	-4.504	6.66e-06	***
yod2010-2022	27.746	< 2e-16	***
sexMale	1.360	0.173743	
eventCancer death	0.191	0.848272	
eventOther death	0.001	0.999476	
raceHispanic (All Races)	3.424	0.000617	***
raceNon-Hispanic American Indian/Alaska Native	0.234	0.815015	
raceNon-Hispanic Asian or Pacific Islander	-0.685	0.493139	
raceNon-Hispanic Black	4.224	2.40e-05	***
raceNon-Hispanic Unknown Race	-0.317	0.750891	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95
age70+	1.525e+00	6.557e-01	1.465e+00
age50-59	1.021e+00	9.791e-01	9.731e-01
age60-69	1.098e+00	9.111e-01	1.050e+00
stageRegional	1.748e+00	5.720e-01	1.700e+00
stageDistant	3.889e+00	2.571e-01	3.754e+00
stageUnstaged	1.466e+00	6.819e-01	1.396e+00
subsiteTrunk	9.946e-01	1.005e+00	9.645e-01
subsiteArm	9.187e-01	1.088e+00	8.879e-01
subsiteLeg	9.213e-01	1.085e+00	8.890e-01
yod2010-2022	1.426e+00	7.013e-01	1.391e+00
sexMale	1.019e+00	9.818e-01	9.919e-01
eventCancer death	3.863e+09	2.588e-10	2.377e-89
eventOther death	1.169e+00	8.557e-01	1.142e-202
raceHispanic (All Races)	1.095e+00	9.136e-01	1.039e+00
raceNon-Hispanic American Indian/Alaska Native	1.024e+00	9.768e-01	8.410e-01
raceNon-Hispanic Asian or Pacific Islander	9.687e-01	1.032e+00	8.846e-01

raceNon-Hispanic Black	1.179e+00	8.485e-01	1.092e+00
raceNon-Hispanic Unknown Race	9.537e-01	1.049e+00	7.118e-01
upper .95			
age70+	1.587e+00		
age50-59	1.072e+00		
age60-69	1.148e+00		
stageRegional	1.798e+00		
stageDistant	4.029e+00		
stageUnstaged	1.540e+00		
subsiteTrunk	1.026e+00		
subsiteArm	9.507e-01		
subsiteLeg	9.548e-01		
yod2010-2022	1.462e+00		
sexMale	1.046e+00		
eventCancer death	6.278e+107		
eventOther death	1.196e+202		
raceHispanic (All Races)	1.153e+00		
raceNon-Hispanic American Indian/Alaska Native	1.246e+00		
raceNon-Hispanic Asian or Pacific Islander	1.061e+00		
raceNon-Hispanic Black	1.272e+00		
raceNon-Hispanic Unknown Race	1.278e+00		

Concordance= 0.979 (se = 0)

Likelihood ratio test= 162841 on 18 df, p=<2e-16

Wald test = 7886 on 18 df, p=<2e-16

Score (logrank) test = 450381 on 18 df, p=<2e-16

```
data_subsite_hn <- data
data_subsite_hn$subsite <- relevel(data$subsite, ref = "Trunk")

# Define a censoring variable at 5 years survival
# By race white
fit.coxph.subsite_hn <- coxph(cancer.death.surv.5yr ~ subsite,
                             data = data_subsite_hn)
summary(fit.coxph.subsite_hn)
```

Call:

```
coxph(formula = cancer.death.surv.5yr ~ subsite, data = data_subsite_hn)
```

n= 415909, number of events= 27536

(1009 observations deleted due to missingness)

	coef	exp(coef)	se(coef)	z	Pr(> z)	
subsiteHead and Neck	0.48210	1.61947	0.01535	31.398	<2e-16	***
subsiteArm	-0.16919	0.84435	0.01774	-9.534	<2e-16	***
subsiteLeg	0.18532	1.20360	0.01779	10.417	<2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

exp(coef) exp(-coef) lower .95 upper .95

subsiteHead and Neck	1.6195	0.6175	1.5715	1.6689
subsiteArm	0.8444	1.1843	0.8155	0.8742
subsiteLeg	1.2036	0.8308	1.1624	1.2463

Concordance= 0.567 (se = 0.002)

Likelihood ratio test= 1680 on 3 df, p=<2e-16

Wald test = 1717 on 3 df, p=<2e-16

Score (logrank) test = 1758 on 3 df, p=<2e-16

```
fit.coxph.subsite_hn_stage <- coxph(cancer.death.surv.5yr ~ subsite + stage,
                                     data = data_subsite_hn)
summary(fit.coxph.subsite_hn_stage)
```

Call:

```
coxph(formula = cancer.death.surv.5yr ~ subsite + stage, data = data_subsite_hn)
```

n= 415909, number of events= 27536

(1009 observations deleted due to missingness)

	coef	exp(coef)	se(coef)	z	Pr(> z)	
subsiteHead and Neck	0.29545	1.34373	0.01542	19.155	< 2e-16	***
subsiteArm	-0.12296	0.88430	0.01775	-6.927	4.3e-12	***
subsiteLeg	0.05410	1.05559	0.01783	3.035	0.00241	**
stageRegional	2.07430	7.95895	0.01402	147.939	< 2e-16	***
stageDistant	3.18260	24.10938	0.01775	179.264	< 2e-16	***
stageUnstaged	0.68300	1.97981	0.02504	27.282	< 2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
subsiteHead and Neck	1.3437	0.74419	1.3037	1.3850
subsiteArm	0.8843	1.13084	0.8541	0.9156
subsiteLeg	1.0556	0.94734	1.0193	1.0931
stageRegional	7.9590	0.12564	7.7432	8.1807
stageDistant	24.1094	0.04148	23.2849	24.9631
stageUnstaged	1.9798	0.50510	1.8850	2.0794

Concordance= 0.752 (se = 0.002)

Likelihood ratio test= 33562 on 6 df, p=<2e-16

Wald test = 44062 on 6 df, p=<2e-16

Score (logrank) test = 78345 on 6 df, p=<2e-16

```
fit.coxph.subsite_hn_full <- coxph(cancer.death.surv.5yr ~ subsite + stage + age + yod + sex + ev
                                     data = data_subsite_hn)
```

Warning in coxph.fit(X, Y, istrat, offset, init, control, weights = weights, :
Loglik converged before variable 12 ; coefficient may be infinite.

```
summary(fit.coxph.subsite_hn_full)
```

Call:

```
coxph(formula = cancer.death.surv.5yr ~ subsite + stage + age +  
      yod + sex + event + race, data = data_subsite_hn)
```

n= 415909, number of events= 27536

(1009 observations deleted due to missingness)

	coef	exp(coef)	se(coef)
subsiteHead and Neck	5.414e-03	1.005e+00	1.570e-02
subsiteArm	-7.933e-02	9.237e-01	1.788e-02
subsiteLeg	-7.656e-02	9.263e-01	1.848e-02
stageRegional	5.586e-01	1.748e+00	1.420e-02
stageDistant	1.358e+00	3.889e+00	1.806e-02
stageUnstaged	3.828e-01	1.466e+00	2.509e-02
age<50	-4.221e-01	6.557e-01	2.042e-02
age50-59	-4.010e-01	6.696e-01	1.884e-02
age60-69	-3.290e-01	7.197e-01	1.580e-02
yod2010-2022	3.549e-01	1.426e+00	1.279e-02
sexMale	1.835e-02	1.019e+00	1.349e-02
eventCancer death	2.207e+01	3.863e+09	1.154e+02
eventOther death	1.558e-01	1.169e+00	2.373e+02
raceHispanic (All Races)	9.038e-02	1.095e+00	2.640e-02
raceNon-Hispanic American Indian/Alaska Native	2.347e-02	1.024e+00	1.003e-01
raceNon-Hispanic Asian or Pacific Islander	-3.176e-02	9.687e-01	4.634e-02
raceNon-Hispanic Black	1.643e-01	1.179e+00	3.889e-02
raceNon-Hispanic Unknown Race	-4.740e-02	9.537e-01	1.493e-01

z Pr(>|z|)

subsiteHead and Neck	0.345	0.730132	
subsiteArm	-4.437	9.14e-06	***
subsiteLeg	-4.142	3.44e-05	***
stageRegional	39.343	< 2e-16	***
stageDistant	75.198	< 2e-16	***
stageUnstaged	15.260	< 2e-16	***
age<50	-20.666	< 2e-16	***
age50-59	-21.289	< 2e-16	***
age60-69	-20.817	< 2e-16	***
yod2010-2022	27.746	< 2e-16	***
sexMale	1.360	0.173743	
eventCancer death	0.191	0.848271	
eventOther death	0.001	0.999476	
raceHispanic (All Races)	3.424	0.000617	***
raceNon-Hispanic American Indian/Alaska Native	0.234	0.815015	
raceNon-Hispanic Asian or Pacific Islander	-0.685	0.493139	
raceNon-Hispanic Black	4.224	2.40e-05	***
raceNon-Hispanic Unknown Race	-0.317	0.750891	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95
subsiteHead and Neck	1.005e+00	9.946e-01	9.750e-01
subsiteArm	9.237e-01	1.083e+00	8.919e-01
subsiteLeg	9.263e-01	1.080e+00	8.933e-01
stageRegional	1.748e+00	5.720e-01	1.700e+00
stageDistant	3.889e+00	2.571e-01	3.754e+00
stageUnstaged	1.466e+00	6.819e-01	1.396e+00
age<50	6.557e-01	1.525e+00	6.300e-01
age50-59	6.696e-01	1.493e+00	6.454e-01
age60-69	7.197e-01	1.390e+00	6.977e-01
yod2010-2022	1.426e+00	7.013e-01	1.391e+00
sexMale	1.019e+00	9.818e-01	9.919e-01
eventCancer death	3.863e+09	2.588e-10	2.382e-89
eventOther death	1.169e+00	8.557e-01	1.143e-202
raceHispanic (All Races)	1.095e+00	9.136e-01	1.039e+00
raceNon-Hispanic American Indian/Alaska Native	1.024e+00	9.768e-01	8.410e-01
raceNon-Hispanic Asian or Pacific Islander	9.687e-01	1.032e+00	8.846e-01
raceNon-Hispanic Black	1.179e+00	8.485e-01	1.092e+00
raceNon-Hispanic Unknown Race	9.537e-01	1.049e+00	7.118e-01
	upper .95		
subsiteHead and Neck	1.037e+00		
subsiteArm	9.567e-01		
subsiteLeg	9.605e-01		
stageRegional	1.798e+00		
stageDistant	4.029e+00		
stageUnstaged	1.540e+00		
age<50	6.825e-01		
age50-59	6.948e-01		
age60-69	7.423e-01		
yod2010-2022	1.462e+00		
sexMale	1.046e+00		
eventCancer death	6.266e+107		
eventOther death	1.195e+202		
raceHispanic (All Races)	1.153e+00		
raceNon-Hispanic American Indian/Alaska Native	1.246e+00		
raceNon-Hispanic Asian or Pacific Islander	1.061e+00		
raceNon-Hispanic Black	1.272e+00		
raceNon-Hispanic Unknown Race	1.278e+00		

Concordance= 0.979 (se = 0)

Likelihood ratio test= 162841 on 18 df, p=<2e-16

Wald test = 7886 on 18 df, p=<2e-16

Score (logrank) test = 450381 on 18 df, p=<2e-16

Censoring the data after 5 years, we compute the Cox proportional hazards models for each covariate in univariable, bivariable (with stage), and multivariable (all covariates) models.

Group	Univariable	Bivariable (with stage)	Multivariable
Age <50 years	0.31 (95%CI: 0.29, 0.32); p<0.001	0.33 (95%CI: 0.32, 0.34); p<0.001	0.66 (95%CI: 0.63, 0.68); p<0.001
Age 50-59 years	0.42 (95%CI: 0.40, 0.43); p<0.001	0.44 (95%CI: 0.42, 0.45); p<0.001	0.67 (95%CI: 0.65, 0.69); p<0.001
Age 60-69 years	0.52 (95%CI: 0.50, 0.53); p<0.001	0.54 (95%CI: 0.52, 0.56); p<0.001	0.72 (95%CI: 0.70, 0.74); p<0.001
Age 70+ years	3.27 (95%CI: 3.15, 3.40); p<0.001	3.04 (95%CI: 2.92, 3.16); p<0.001	1.53 (95%CI: 1.45, 1.59); p<0.001
Men	1.65 (95%CI: 1.60, 1.69); p<0.001	1.49 (95%CI: 1.46, 1.53); p<0.001	0.98 (95%CI: 0.96, 1.01); p = 0.173743
Women	0.61 (95%CI: 0.59, 0.62); p<0.001	0.67 (95%CI: 0.65, 0.69); p<0.001	1.49 (95%CI: 1.46, 1.53); p<0.001
Black	2.07 (95%CI: 1.92, 2.23); p<0.001	1.34 (95%CI: 1.24, 1.45); p<0.001	1.18 (95%CI: 1.09, 1.27); p<0.001
White	0.48 (95%CI: 0.45, 0.53); p<0.001	0.75 (95%CI: 0.69, 0.80); p<0.001	0.85 (95%CI: 0.79, 0.92); p<0.001
Head and Neck	1.62 (95%CI: 1.57, 1.67); p<0.001	1.34 (95%CI: 1.30, 1.39); p<0.001	1.00 (95%CI: 0.98, 1.04); p = 0.730132
Trunk	0.62 (95%CI: 0.60, 0.64); p<0.001	0.74 (95%CI: 0.72, 0.77); p<0.001	0.99 (95%CI: 0.96, 1.03); p = 0.730132
Arm	0.52 (95%CI: 0.50, 0.54); p<0.001	0.66 (95%CI: 0.64, 0.68); p<0.001	0.92 (95%CI: 0.89, 0.95); p<0.001
Leg	0.74 (95%CI: 0.72, 0.77); p<0.001	0.79 (95%CI: 0.76, 0.81); p<0.001	0.92 (95%CI: 0.79, 0.95); p<0.001